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A smart printing service for students at HCMUT

Group CC01-03

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Revision History

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All member	04/11/2024 - 16/12/2024	Implementation sprint 2	Version 4

1. Task 1: Requirement elicitation

1.1 Domain Context

To enhance the working and studying experience on campus, Ho Chi Minh City University of Technology (HCMUT) decides to build an app called the Student Smart Printing Service (HCMUT_SSPS) to provide document printing service for students. This system can be accessed by all students using the student authentication service (HCMUT_SSO).

Based on school privacy, each account is given a predetermined number of pages per semester for printing. Students can purchase additional pages through the university's payment service to increase their pages limit. Once logged in, students can view detailed information about available printers (including printer ID, location on campus,...) and upload documents for printing. They are required to type in the number of pages, color, size, and schedule the print time. However, printing is restricted if the page limit is exceeded.

The system also provides an administrator configuration for the Student Printing Service Officer (SPSO) to manage the printers, configure settings, and keep track of the transactions. Additionally, the system needs to offer a user-friendly interface to be available for all and generate periodic reports in a customizable format for the SPSO.

1.2 Stakeholders and Needs

Students: Students in the HCMUT campus are the main users of the HCMUT_SSPS system. As a user, they need a convenient web-based app or mobile app for printing their documents through their student account. For each semester, they are given a number of pages for printing, they can also buy more pages if needed. All students are required to view their printing and payment history.

Student Printing Service Officer (SPSO): SPSO requires some features to set constraints related to file type, keep track of students' printing process, and ensure the system has sufficient resources (printers, paper, etc.) for the students.

IT staff: They are responsible for developing, maintaining, and ensuring the smooth operation of the HCMUT_SSPS system. This includes managing system security, and

integration with the sign-in and payment system of the university. The IT staff and ensuring the service works on both web and mobile platforms.

Online Payment Service Administrator (OPSA): They handle the financial transactions related to the purchase of additional printing pages. Their role includes ensuring secure, smooth payments through systems such as BKPay and maintaining records of these transactions.

Ho Chi Minh City University of Technology (HCMUT): HCMUT is an organization that licenses and invests in Student Smart Printing Service to enhance student convenience by providing an online system for document printing.

1.3 Benefits of the System

With the idea of this project, the Student Smart Printing Service, we tend to offer multiple benefits for both HCMUT's students and SPSO. One of the primary advantages is students now do not have to visit photocopy stalls outside the university campus to print their document. This can also prevent the risk of plagiarism since some photocopy stalls may sneak and sell one student's report or document after it is printed to other students. Nowadays, students are getting used to online transactions so with the combination of the BK pay system for prepayment, they can easily manage the feasible balance. With the support of the system, the SPSO can control the access traffic of students and also manage printers remotely most of the time efficiently.

The most significant improvement of this system compared to normal printing service is that SPSO can improve the service through monthly reports and printing logs of students. Based on these statistics and insights, the SPSO can simply enhance the service by increasing the default page balance of each student, or upgrade the printing system in order to meet the needs of all the users in the HCMUT campus. Furthermore, SPSO can also create an internship chance (for example managing, and optimizing the system) for CSE students from HCMUT, giving them more chances to encounter real-life projects, smart printing service in this case.

In summary, this project aims to offer a smart, secure, optimized printing service for students in HCMUT. It not only improves the daily experience of students but also can give some of them a career change in this field.

1.4 Functional Requirements

- **Student:**

- The system allows students to print a document by uploading a document (when it does not exceed his/her account balance) onto the system.
- The system allows students to choose a printer, and specify the printing properties: paper size, pages (of the file) to be printed, one-/double-sided, number of copies, etc.
- The system allows students to view his/her printing log for a time period together with a summary of the number of printed pages for each page size.
- The system allows students to buy more pages using the feature Buy Printing Pages of the system and pay the amount through the online payment BKPay system.
- The system provides a support feature that allows students to provide feedback or complaints about the system.
- The system sends real-time notifications to students when their documents are completely printed.
- The system requires student confirmation before printing or purchasing additional pages. Once confirmed, students will not be able to cancel the tasks.
- The system automatically backs up all printing and transaction history to prevent student data loss due to system failures or security breaches.

- **SPSO:**

- The system allows SPSO to manage the configuration of the system such as changing the default number of pages, the dates that the system will give the

default number of pages to all students, and the permitted file types accepted by the system.

- The system allows SPSO to add/enable/disable a printer.
- The reports of the use of the printing system are generated automatically at the end of each month and each year are stored in the system and can be viewed by the SPSO.
- The system allows SPSO to monitor and check all student authentication activities. It should provide detailed logs of authentication attempts, including timestamps, IP addresses, and success/failure status.
- The system will automatically send a notification to the SPSO whenever a printer runs out of paper or toner.
- The system stores all student feedback in a repository, which is accessible to SPSO.
- The system allows SPSO to restore student data from backups in case of data loss or corruption.

- **IT staff:**

- The system allows IT staff to configure and update printer settings, specifications, policies, and user guidelines.
- When a technical error occurs on a system or printer, the system will automatically send a notification to the IT staff.
- IT staff are allowed to disable all of the system.
- IT staff can customize notes, warnings, and notifications related to printer usage. These messages can be displayed directly on the printer's user interface or sent to users via email.
- The system will store a history of occurred errors for analysis and system improvement.

- **Online Payment Service Administrator (OPSA):**

- The administrator can adjust the features of the Buy Printing Pages of the system, such as price or purchase limit.
- The administrator can oversee the processing of payment transactions, including authorization, settlement, and reconciliation.
- The system generates reports on payment transactions, revenue, and expenses related to the printing service, and can be viewed by the administrator.

- **University manager:**

- A detailed report of the operational costs of the system, including the cost of paper, ink, printer maintenance, and other related expenses is generated automatically at the end of each month and is stored in the system, and can be viewed by the manager.
- The system stores all student feedback in a repository, which is accessible to the manager.

1.5 Non- Functional Requirements

- **Performance: (Response Time and Throughput)**

- **Response Time:** The service's response time to user actions, such as receiving a print request, verifying the document, and placing it in line for printing. In times of high demand, this prompt reaction guarantees a flawless user experience by reducing waiting times and student discomfort. The service's response time should be optimized *from 5 to under 2 seconds based on the size of the files*. According to Google, a website responds well when the TTFB is under 200 milliseconds. This prompt reaction ensures a seamless user experience by minimizing waiting times and reducing potential discomfort for students, especially during peak demand periods. Quick processing is essential to maintain a smooth and efficient workflow, ensuring that users can rely on the service even when demand is high.

- **Throughput:** The printing service must be capable of handling high user loads, particularly during peak times such as exams and project deadlines. The system should scale to support *at least 100 simultaneous users* without any noticeable performance degradation. Each print job, from document submission to queue placement, should be processed within *3 to 5 seconds*, even under heavy demand. This ensures students can rely on the system for timely and efficient service during critical moments. Effective load balancing and optimized resource allocation will be implemented to maintain this performance standard.
- **Security (User Authentication, Data Encryption, and Auto Maintenance):**
 - **User Authentication:** This service is open to everyone. Thus, security must be one of the most considered aspects. The system must integrate with the *university's single sign-on (SSO) system to authenticate users*. This ensures that only authorized students and staff can access the service, and students don't need to remember additional qualifications.
 - **Data Encryption:** Because of the project files will be uploaded to this service. So that *all personal data must be encrypted* both in transit and at rest. This includes student credentials, documents, and payment information (if applicable). Data encryption ensures that no one, except authorized users, can access confidential files.
 - **Auto maintenance:** The system will incorporate automatic maintenance routines, such as clearing unused memory and removing outdated backup files, after a specified period. These tasks will be scheduled to run during off-peak hours to minimize service disruption. For example, *memory caches will be automatically cleared every 24 hours* to prevent memory overflow and potential performance issues. *Old backup files will be removed after 30 days*, ensuring optimal storage usage while maintaining security. *Regular automated checks will also be performed to identify and remove potential*

security threats, such as unauthorized files or vulnerabilities, helping to maintain a secure and efficient system.

- **Availability & Reliability (Uptime and Failure Recovery):**

- **Availability:**

- To minimize service disruptions, the system must achieve a 99.9% uptime. This means that, excluding scheduled maintenance, the system can only experience up to approximately **8 hours** of downtime annually—roughly equivalent to a full day of study or work.

- Considering the actual learning and working hours at HCMUT, we have determined the *service's operational hours to be from 7:00 am to 6:00 pm*. Ensuring high availability during these hours is crucial, especially during key moments like deadlines or exams, where students need reliable access to the system.

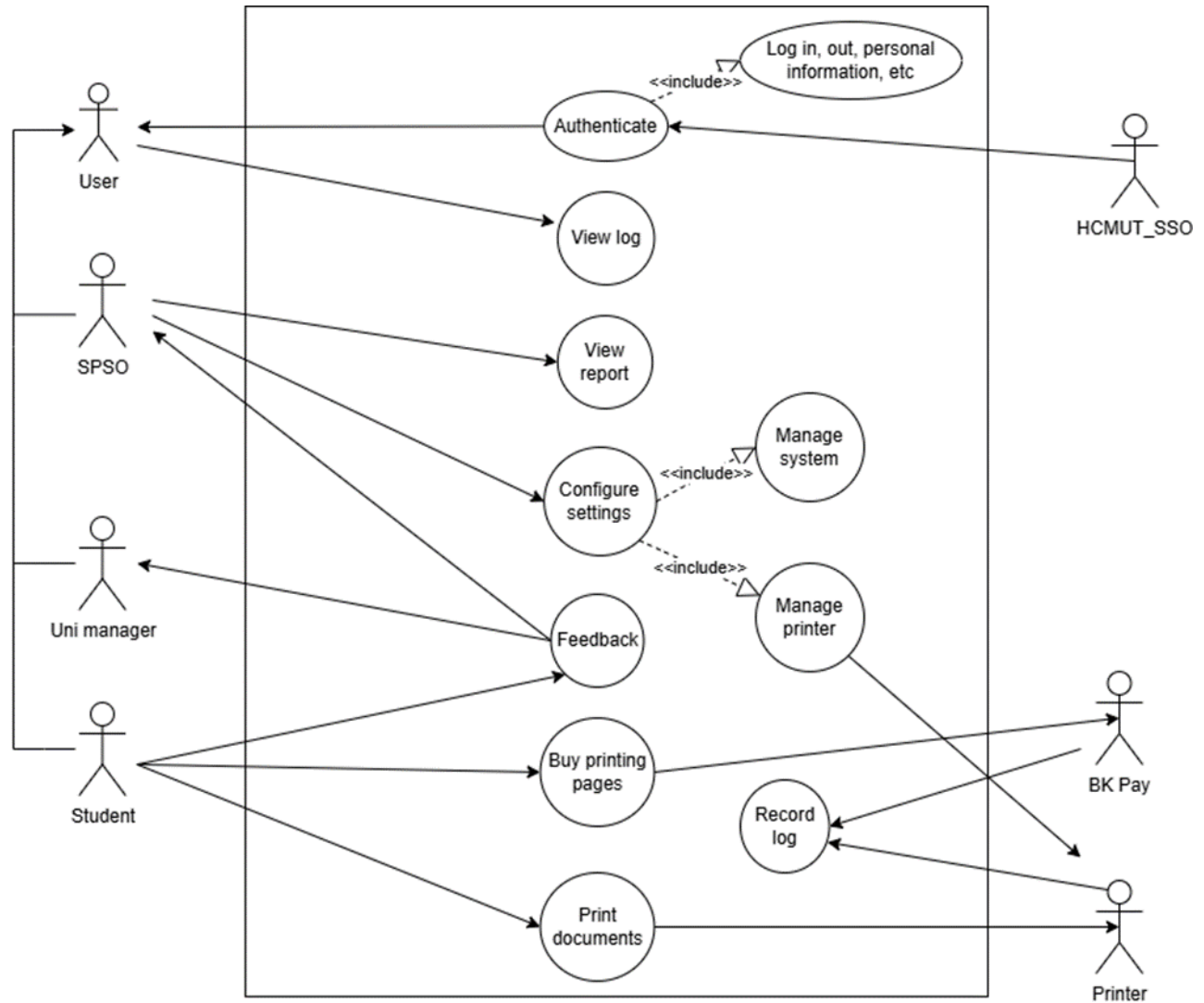
- **Reliability (Failure Recovery):** To minimize disturbance in the event of a printer failure, the system ought to automatically reroute print jobs to the closest accessible printer without requiring user intervention. Users must be informed as soon as possible if the entire printing network is unavailable, together with an estimated time of recovery. Failures and recovery measures need to be tracked via a reliable logging system for analysis and future enhancements. The system should also protect against data loss during malfunctions, enabling print jobs to continue without interruption after the problem is fixed.

- **Changing languages:** The system will support multiple languages, including *English and Vietnamese*. Students can choose their preferred language at the login screen or switch languages in their settings. This ensures accessibility for both local users and international students or faculty members at HCMUT.

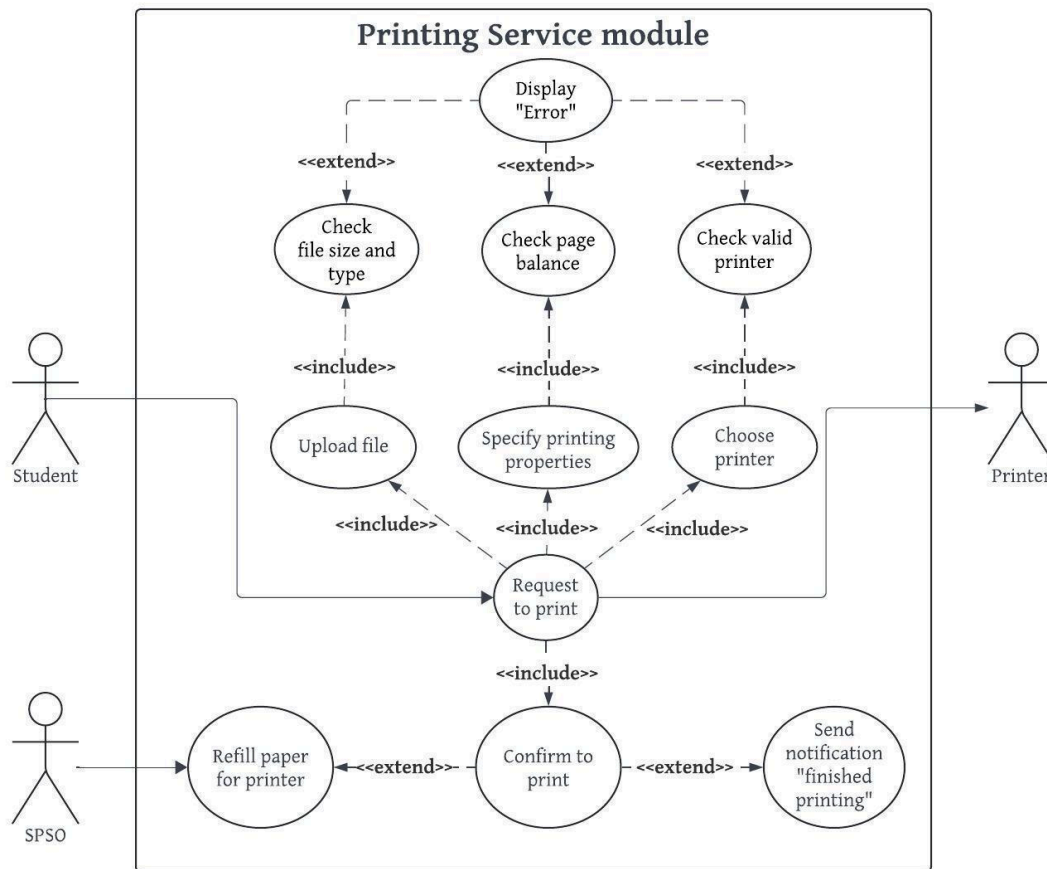
- **Graphical User Interface (GUI):** The user interface is designed to be *simple, intuitive, and highly efficient*. Key features like file uploading, monitoring progress, and customizing print settings are easily accessible from the main screen. The drag-and-drop file upload functionality, real-time feedback via progress bars, and clear error messages ensure users experience minimal friction while completing tasks. Interactive elements such as customizable print options (e.g., color settings, paper size) are laid out with live previews, reducing the number of clicks needed to finalize tasks.
- **File size limit:** To ensure smooth operation and efficient processing, the system will impose a *maximum file upload limit of 50 MB per document*. Larger files may need to be compressed or split into smaller sections before upload. This limitation helps maintain system performance, especially during periods of high demand.
- **Permitted file type:** The system just allow student to upload documents with file extensions are *pdf, doc or docx*

1.6 Use-case Diagrams

1.6.1 Use-case Diagram for the Whole System



1.6.2 Use-case Diagram for Printing Service Module



1.6.3 The Details of Usecases in Printing Service Module

a. Usecase “Upload File”

ID and Name	UC-1 Upload file
Created by	Lê Hà Nguyên Khánh
Date created	30/09/2024
Actors	Students, Printer
Description	The student uploads their document file to the system, then the system checks the file size and file type (file extension)
Trigger	The student selects the “Print” option in the HomeUI and enters the Upload interface.
Preconditions	PRE-1. The system is not in annual maintenance PRE-2. The system is online PRE-3. Student logged into the system

	PRE-4. Student's device is connected to the Internet PRE-5. Student is authorized to upload documents PRE-6. The file to be uploaded is accessible
Postconditions	POST-1. The file is uploaded to the system
Normal Flow	1. Student upload their file to the Upload interface 2. The system checks whether the file size is valid 3. If the file size is valid, then the system checks whether the file type is valid 4. If the file type is valid, the Upload interface displays the message "Valid file". 5. The system stores the file into a buffer. 6. The student presses "Next" and then enters the Specify interface.
Alternative Flows	None
Exceptions	The file size is invalid 3a. If the file exceeds 50MB, the Upload interface displays "File is exceeding 50MB" The file type is invalid 4a. If the file type is invalid, the Upload interface displays "Invalid file type" (Then return to Step 1 in Normal Flow)

b. Usecase "Specify Printing Properties"

ID and Name	UC-2 Specify printing properties
Created by	Hoàng Văn Phi
Date created	30/09/2024
Actors	Students, Printer
Description	The student specifies the printing properties, then the system checks the page balance of the student
Trigger	The student enters the Specify interface after uploading their document file
Preconditions	PRE-1. The system is not in annual maintenance PRE-2. The system is online PRE-3. Student logged into the system PRE-4. Student's device is connected to the Internet PRE-5. Student had already upload the valid document PRE-6. Student is authorized to specify printing properties

Postconditions	POST-1. The user finish specifying properties
Normal Flow	1. The system displays a list of printing properties 2. Student specifies printing properties: • Paper size • Pages (of the file) to be printed • One-/double-sided • Number of copies 3. Student press “OK” 4. The system calculates page consumption for the document 5. The system retrieves the user’s latest page balance. 6. The system checks whether the page balance is enough 7. If the page balance is enough, the Specify Interface will display the message “Valid properties” 8. System store the specified properties 9. The student presses “Next” and then enters the ChoosePrinter interface.
Alternative Flows	The page balance is not enough 7a. If the page balance is not enough, the Specify Interface will prompt user will prompt user to change properties or not: 8a. If the user selects “Yes”, the Specify Interface will prompt user to specify again or purchase more pages • If the user selects “Specify again”, the Specify Interface returns to the initial state (Back to step 2 in normal flow). • If the user selects “Buy more pages” , the system will navigate to BK PAY interface.
Exceptions	The user selects “No” 8b. If the user selects “No”, the Specify Interface will return to the HomeUI.

c. Usecase “Choose Printer”

ID and Name	UC-3 Choose printer
Created by	Huỳnh Mai Quốc Khang
Date created	30/09/2024
Actors	Student, Printer
Description	Student chooses one of the printers from the system for printing documents
Trigger	The student enters the ChoosePrinter interface after specifying the properties

Preconditions	PRE-1. The system is not in annual maintenance PRE-2. The system is online PRE-3. Student logged into the system PRE-4. The student's device is connected to the Internet PRE-5. The student is authorized to print documents PRE-6. The student had already specified properties
Postconditions	POST-1. A printer is successfully selected and system pushes a request to print-queue
Normal Flow	Select a printer from the list 1. The ChoosePrinter interface displays a list of printers, including their details and status. 2. The student selects one of the printers in the list. 3. The system checks whether the printer is suitable for printing 4. If the printer is available, the system checks whether the selected printer has enough paper for printing 5. If the printer has enough paper, the ChoosePrinter Interface will display the message "Valid printer" 6. The student presses "Print" and then the system pushes a request to print-queue.
Alternative Flows	Select a printer by its ID 1a. The system provides a search field where the student can enter the printer ID to find a specific printer. 2a. The student enters the printer ID in the field. 3a. The system searches for the printer. If that printer is found, the interface displays it along with its details and current status. If the printer is not found, the interface displays a blank list. (Back to step 1 in normal flow if the student removes the ID). The printer is unavailable 4b. If the printer is unavailable, the ChoosePrinter Interface will display the message "Invalid printer" and return to the printer lists (Then return to step 1 in normal flow). The printer does not have enough paper 5c. If the printer does not have enough paper, the Choose Interface displays a notification "You will take more time for printing" and prompts the user to select printer again or continue: • If the student chooses "Continue", the system pushes a request to print-queue.

	If the student chooses “Select again”, the Choose Interface returns to the printer lists (Then return to step 1 in normal flow).
Exceptions	5a. If the printer is invalid, the system will display a “This printer is unavailable” message with the detailed reason.

d. Usecase “Confirm to Print”

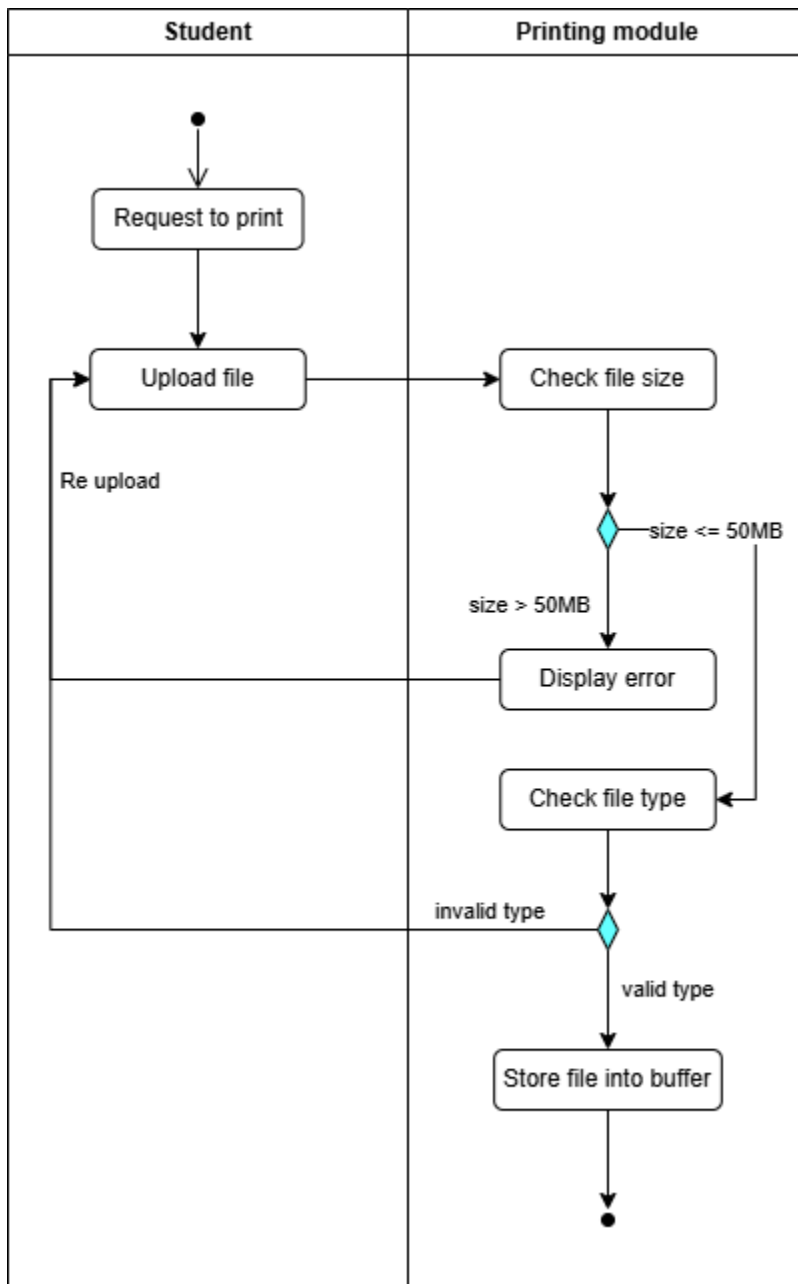
ID and Name	UC-4 Confirm to print
Created by	Huỳnh Ngọc Văn
Date created	30/09/2024
Actors	Student, SPSO
Description	After request to print the uploaded document, the system start to print the document file
Trigger	Student select the “Confirm to print” option in the system’s interface
Preconditions	PRE-1. The system is not in annual maintenance PRE-2. The system is online PRE-3. Student logged into the system PRE-4. Student’s device is connected to the Internet PRE-5. Student had already upload the valid document
Postconditions	POST-1. The document is printed successfully
Normal Flow	1. The system retrieves the first print request in print-queue 2. The system checks if the printer’s remaining paper is enough 3. If the remaining paper is enough, the printer starts printing 4. The system updates the printing status and saves the log 5. When finishing printing, the system sends a “Finished Printing” notification to the student.
Alternative Flows	If the printer’s remaining paper is not enough 3a. If the remaining page is not enough, the system pushes that printing request to wait-queue 3a1. The system sends a “Printer is needed refilled paper” notification to SPSO 3a2. SPSO refills paper and clicks the “Complete” button on the printer’s dashboard. 3a3. The system retrieves the first request from the wait-queue and pushes it to the head of the print-queue. (Then return to step 1 in normal flow)

Exceptions	None
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2. Task 2: System Modeling

2.1 Activity Diagram

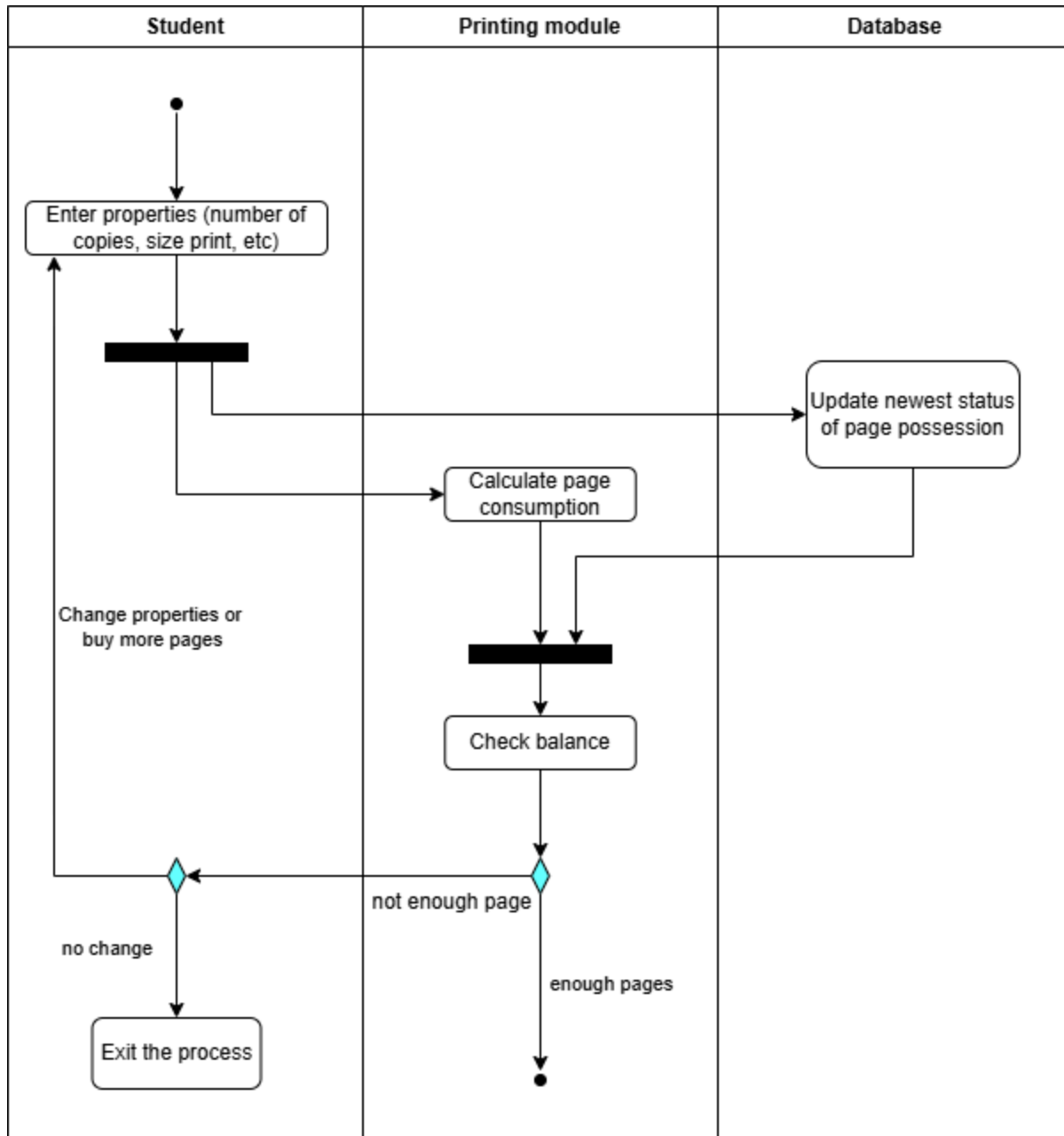
2.1.1 Uploading file



Description of uploading file activity diagram:

1. Student after login to the system can enter the printing service by sending a request.
2. First, student clicks the button “Upload file” to upload the documents that need to be printed.
3. After receiving the files from the student, the printing module starts to check the size of the file.
 - a. If the size is greater than 50MB, display the error to the student and then request the student to return to step 2 to upload another file.
 - b. On the other hand, if the size is no greater than 50MB, then start the next step to check the type of documents.
4. Check the acceptable file type in a list of types that SPSO provides.
 - a. If the type is invalid then display error and let student return to step 2 to upload another file.
 - b. If the type is valid then pop-up “Upload successfully”
5. Store the file into request buffer and finish checking the format.

2.1.2 Specify printing properties

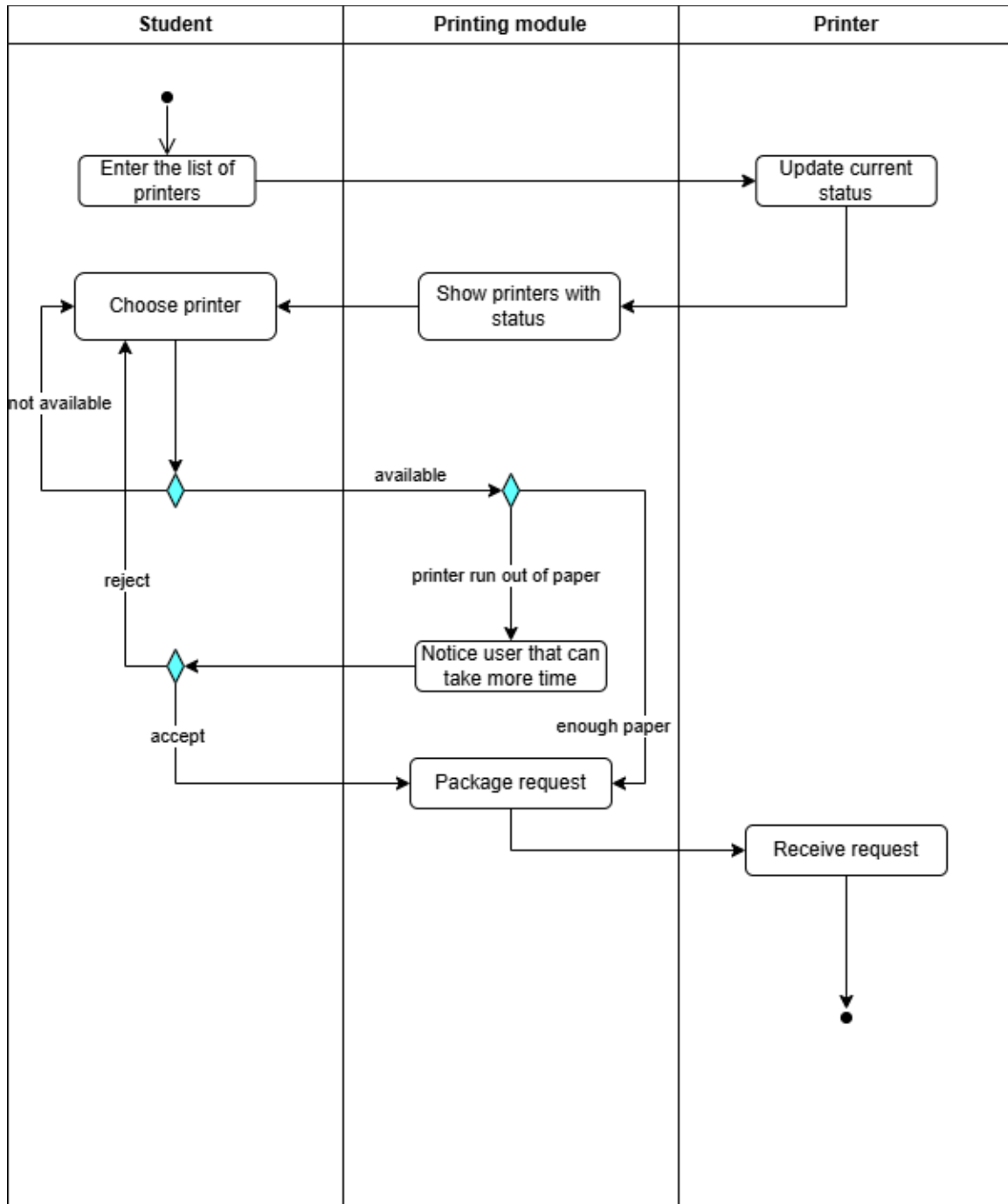


Description of specify printing properties activity diagram:

1. After checking file format, student enter all of the printing properties such as:
 - a. Paper size (A4, A3,...)
 - b. Pages to be printed
 - c. One-sided or double-sided paper
 - d. Number of copies
2. System then calculates the number of pages consumed and displays it to student.

3. System checks the newest status of page possession from student account.
4. Check if the page consumption exceeds the page own.
 - a. If there are enough pages then finish this step.
 - b. If there is not enough page
 - i. Exit the process if student chooses nothing to change.
 - ii. Else if they want to change properties or buy more pages, then return to the starting point.

2.1.3 Choose printer

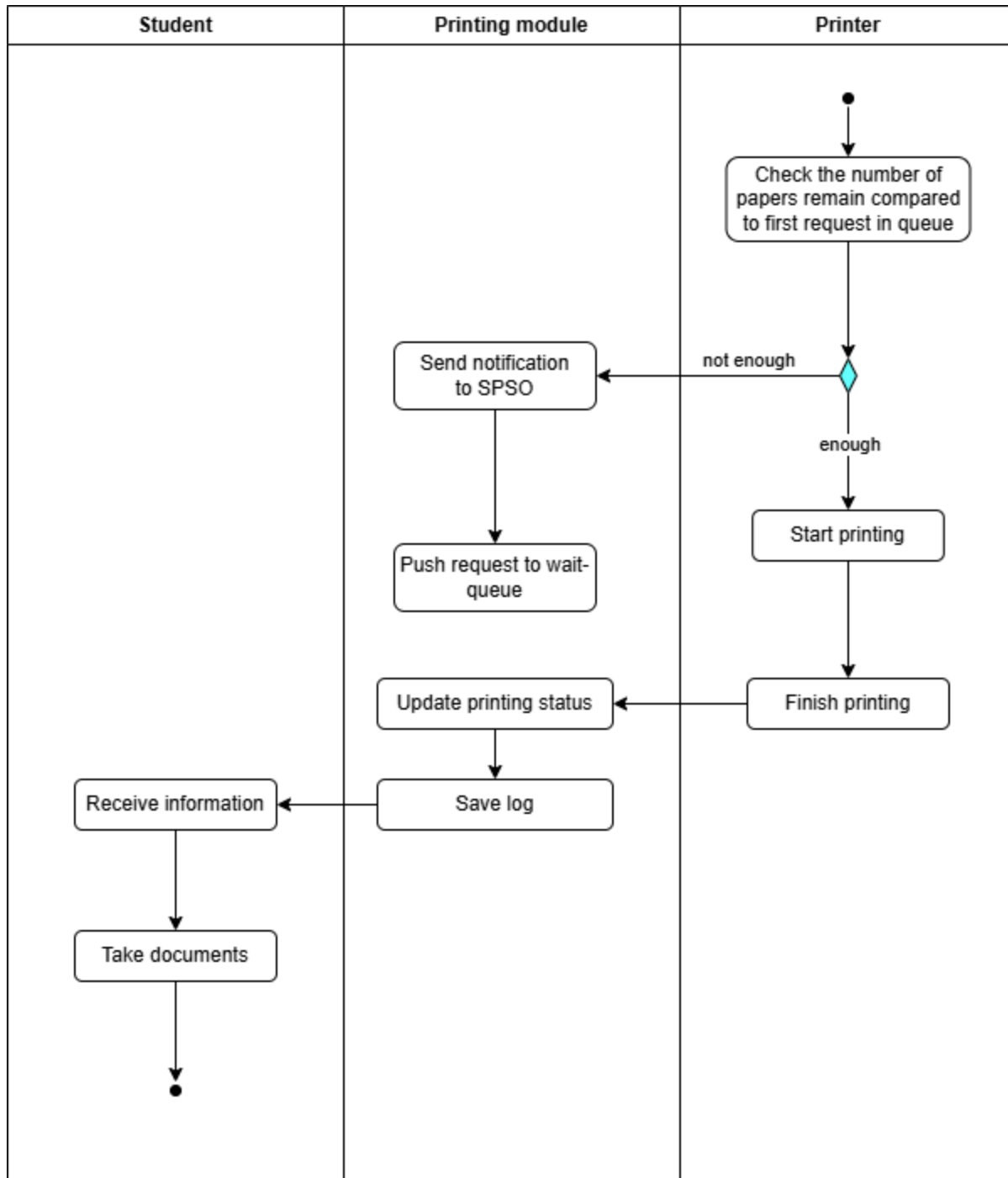


Description of choosing printer activity diagram:

1. After entering properties, the list of printers will be displayed for student to choose.
2. Printers will update the newest status (available or not) for the printing module.
3. Student choose a specific printer
 - a. If the printer is available then go to the next step.

- b. If it is not available, then choose another printer.
- 4. Check if the chosen printer has run out of paper or not.
 - a. If it does not have enough paper, then notify the user that it may take more time.
 - i. If user chooses “Okay”, then move to the next step.
 - ii. Else then go back to step 3 to choose another printer.
 - b. If it has enough paper, move to step 5.
- 5. Package the request (include all information such as properties, files, printer) then send it to printer controller.

2.1.4 Printing processing

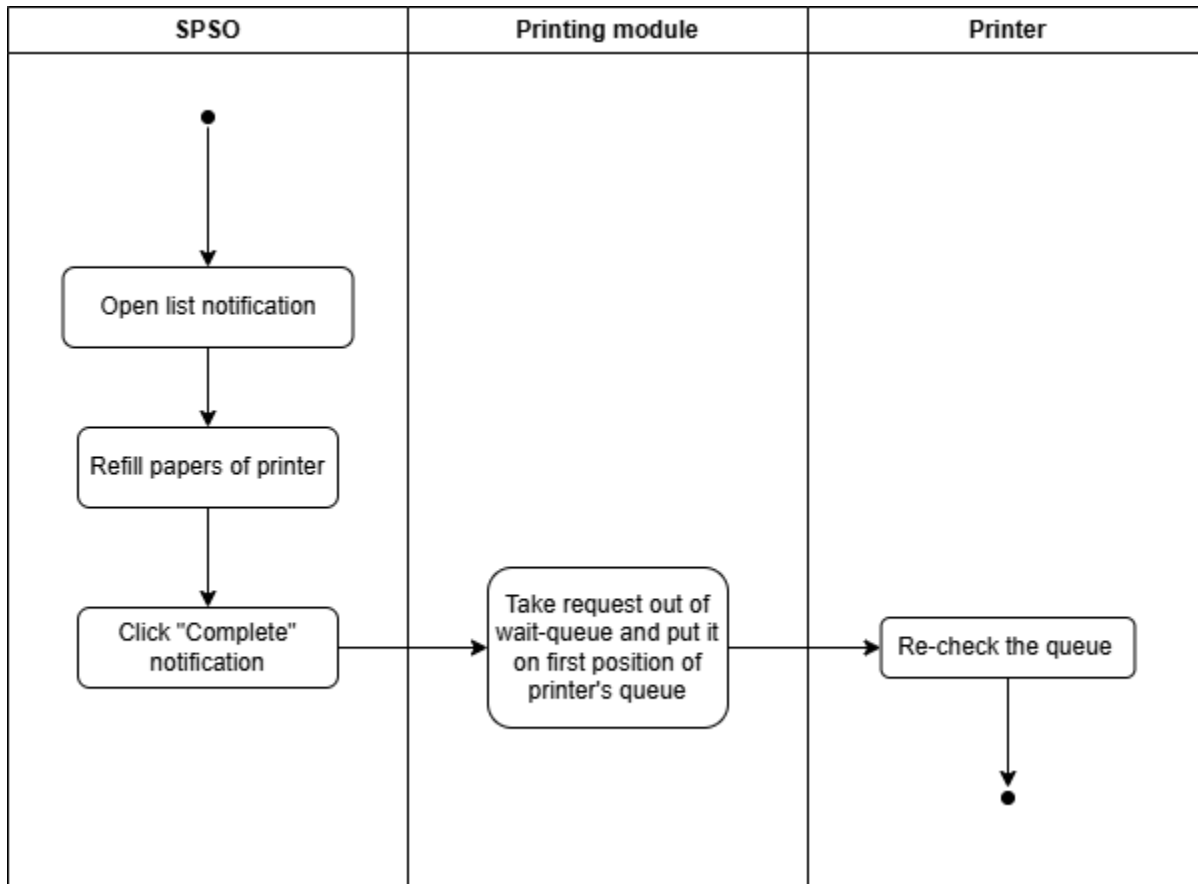


Description of print processing activity diagram:

1. Printer controller checks the running-queue to take the first request in the queue.
2. Printer then checks if the amount of papers in the printer is enough to process that request or not.

- a. If it is not enough then send a notification to SPSO and push that request to the wait-queue in Printing module.
 - b. On the other hand, the process continues.
3. When the printer finishes printing, the system will update the status and notify student to take the documents.

2.1.5 Refill paper

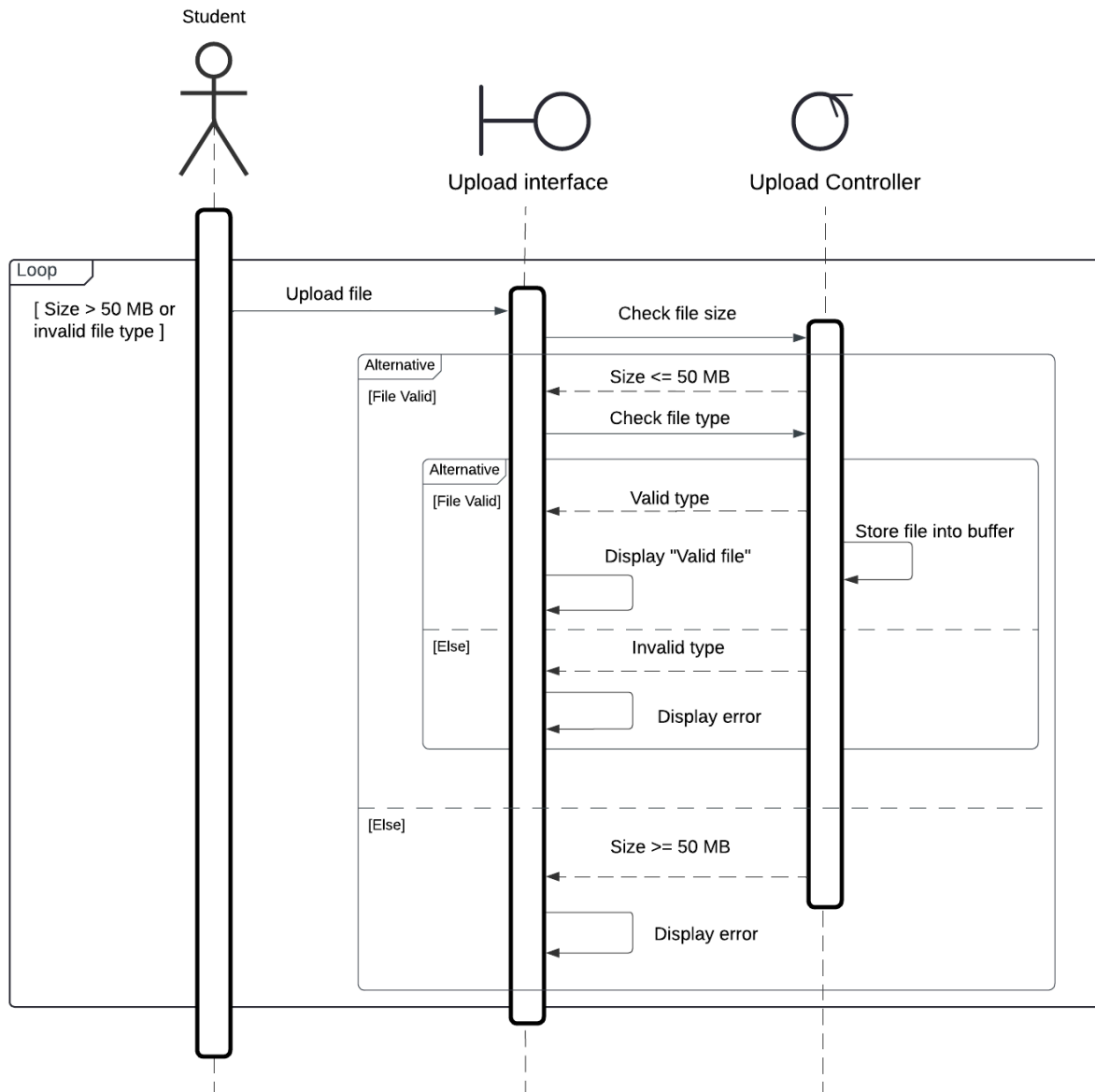


Description of refill paper process:

1. SPSO checks the list of notifications and goes to refill the paper.
2. After finishing, SPSO will click “Complete” on that notification request.
3. The printing module then takes the request out of the wait-queue and puts it to the first position in the running-queue to ensure fairness.
4. The printer starts print processing activity.

2.2 Sequence Diagram

2.2.1 Upload file

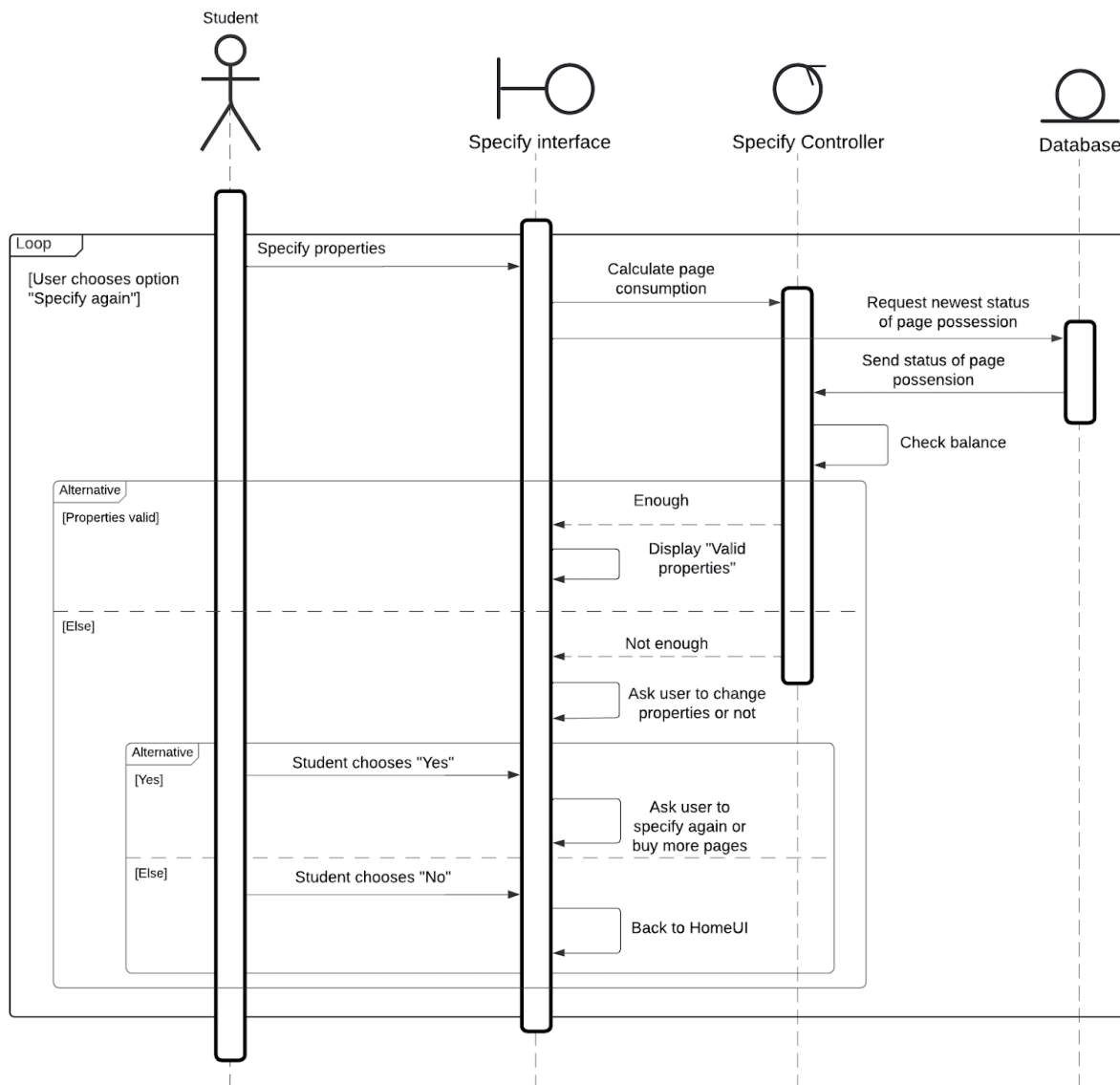


Describes the steps to perform the “Upload File” Task from the Sequence Diagram:

1. The user selects the option “Upload file” in the HomeUI and enters the Upload interface.
2. The user uploads a file, and the Upload interface then calls the `checkFileSize()` function in the Upload Controller. The Upload Controller will check if its size is valid and return a message (Valid/Invalid) to the Upload Interface:

- a. If the size is valid, the process continues.
 - b. If the size is invalid, the Upload Interface will display a message “Invalid file” (Back to step 2).
3. The Upload interface calls the checkFileType() function in the Upload Controller. The Upload Controller will check if its type is valid and return a message (Valid/Invalid) to the Upload Interface:
 - a. If the type is valid, the Upload Controller stores the document file into a buffer and the Upload Interface will display the message “Valid file” then the user presses “Next” (Proceeding to the “Specify properties” task).
 - b. If the type is invalid, the Upload Interface will display an error message “Invalid file” (Back to step 2).

2.2.2 Specify properties

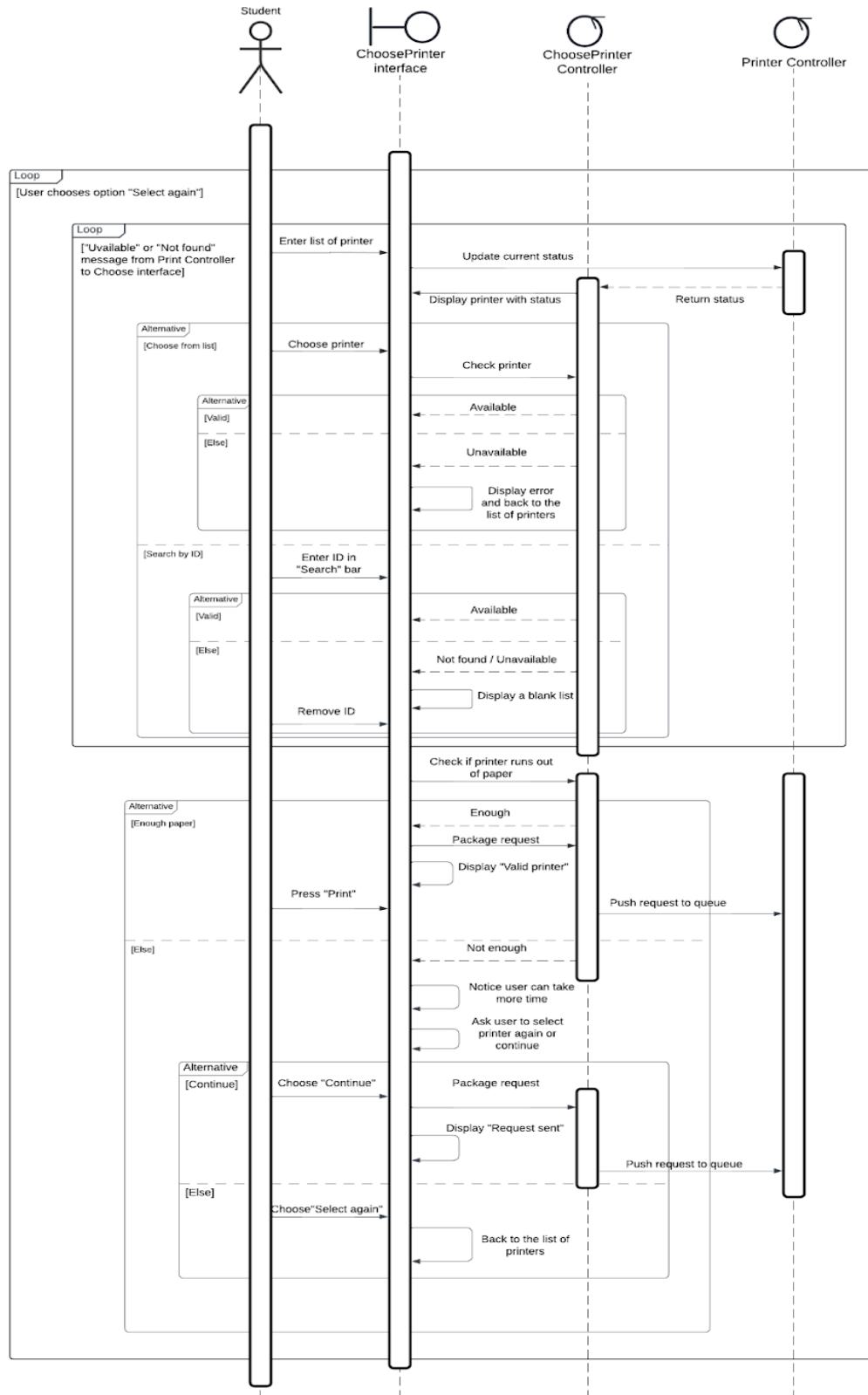


Describes the steps to perform the "Specify properties" Task from the Sequence Diagram:

1. After uploading a document, user enters the Specify Interface.
2. In the Specify interface, user specifies printing properties including Paper size, pages (of the file) to be printed, one-/double-sided printing, number of copies.
3. The Specify Interface calls the page calculating task in the Specify Controller to calculate the number of total page consumption for the document.
4. The Specify Interface sends a request to the Database to retrieve the latest page balance, which is then sent to the Specify Controller.

5. The Specify Controller checks whether the user's page balance is sufficient and returns a message (Enough/Not enough) to the Specify Interface:
 - a. If the page balance is enough, the Specify Interface will display a message "Invalid properties" then the user presses "Next" (Proceeding to the "Choose Printer" task).
 - b. If the page balance is not enough, the Specify Interface will prompt user to change properties or not:
 - i. If the user selects "Yes", the Specify Interface will prompt user to specify again or purchase more pages:
 - If the user selects "Specify again", the Specify Interface returns to the initial state (Back to step 2).
 - If the user selects "Buy more pages" , the system will navigate to BK PAY interface.
 - ii. If the user selects "No", the Specify Interface will return to the HomeUI.

2.2.3 Choose printer

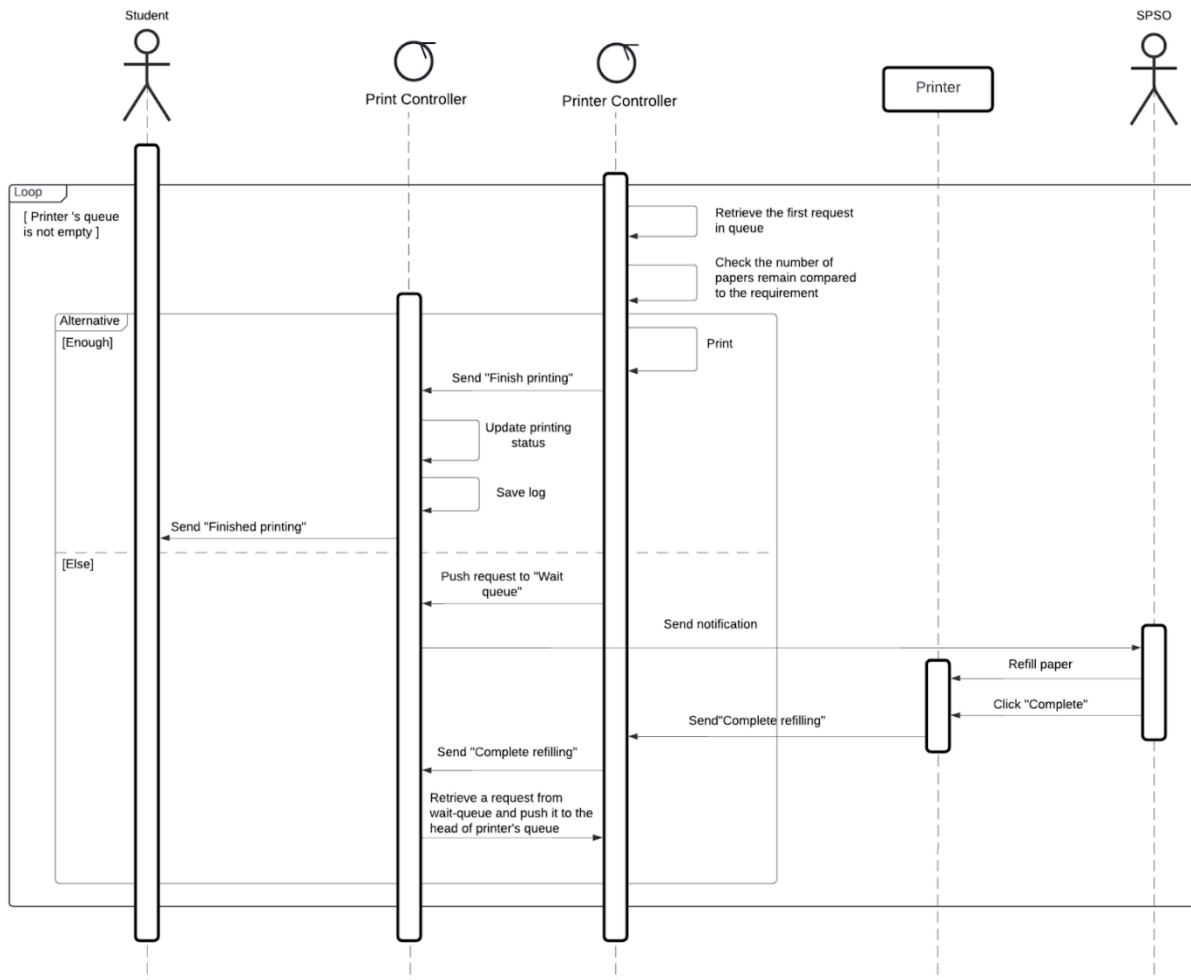


Describes the steps to perform the “Choose Printer” task from the Sequence Diagram:

1. After specifying the printing properties, user selects option “Choose a Printer” in the HomeUI and then enters the Choose Interface.
2. The Choose Interface calls the updating status task in the ChoosePrinter Controller, which returns the current status of printers to the ChoosePrinter Controller
3. The ChoosePrinter Controller then calls a display task to display the printers list with their status in the Choose interface.
4. Student starts selecting printer:
 - a. If the user selects a printer from the list, the Choose Interface will call the selectPrinter() function in the ChoosePrinter Controller. The ChoosePrinter Controller will check its status and return a message (available/unavailable) to the Choose Interface:
 - i. If the printer is available, the process continues.
 - ii. If the printer is unavailable, the ChoosePrinter Interface displays “Invalid printer” and return to the printer lists (Back to step 2).
 - b. If the user searches for a printer by ID, the Choose Interface will call the searchByID() function in the ChoosePrinter Controller. The ChoosePrinter Controller will check if the printer exists and its status, then return a message (Available/Unavailable/Not found) to the Choose Interface:
 - i. If the printer is available, the process continues.
 - ii. If the printer is not found or unavailable, the Choose Interface will display a blank list and then the student removes the ID (Back to step 2).
5. The Choose Interface will call the checking paper task in the ChoosePrinter Controller, the ChoosePrinter Controller checks whether the selected printer has enough paper for printing, then returns a message (Enough/Not enough) to the Choose Interface:
 - a. If the printer has enough paper, the Choose Interface displays “Valid printer”.

- i. The student presses “Print” and then the ChoosePrinter Controller will push a request to the print-queue of Printer Controller (Proceeding to the “Print” task).
- b. If the printer has not enough paper, the Choose Interface displays a notification “You will take more time for printing” and prompts the user to select printer again or continue:
 - i. If the user chooses “Continue” option, the ChoosePrinter Controller will push a request to print-queue of Printer Controller (Proceeding to the “Print” task).
 - ii. If the user chooses “Select again” option, the Choose Interface returns to the printer lists (Back to step 2).

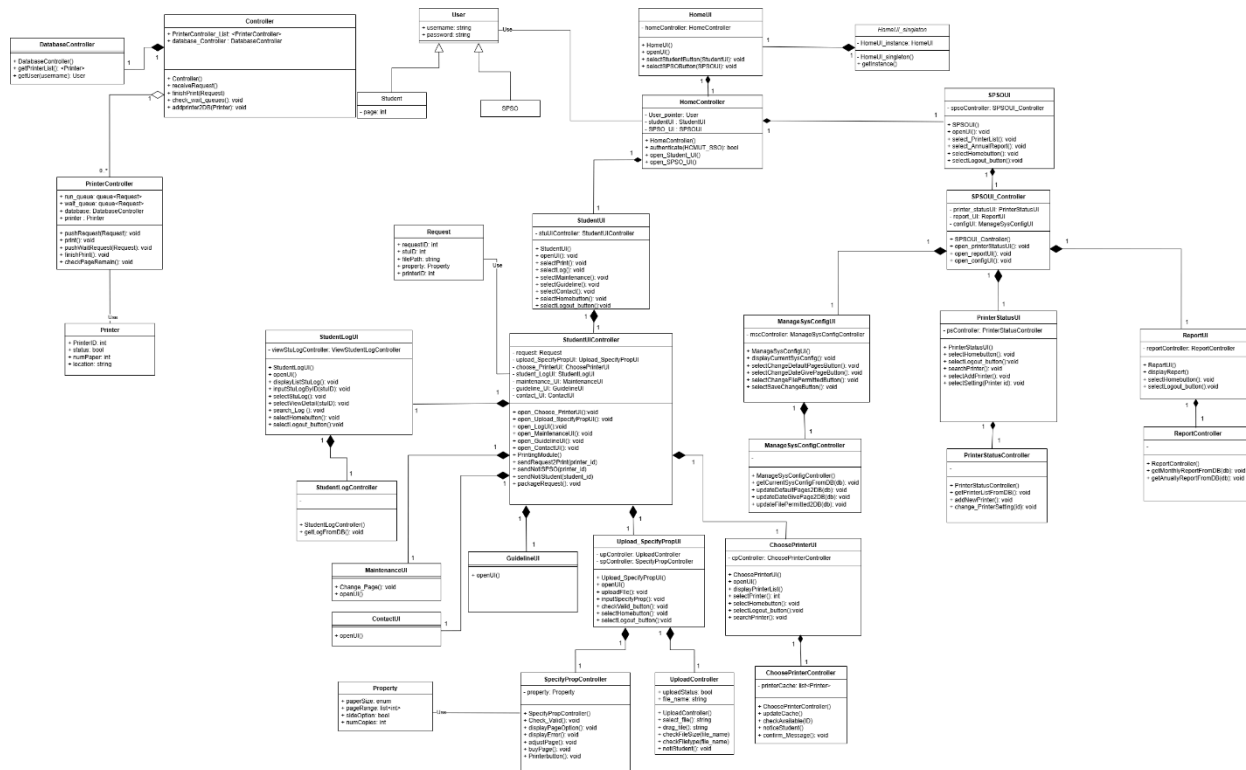
2.2.4 Print and refill paper



Describes the steps to perform the “Print ” and “Refill paper” tasks from the Sequence Diagram:

1. The Printer Controller retrieves the first request in the print-queue.
2. The Printer Controller checks the remaining paper and compares it to the number of papers required for the request:
 - a. If the number of paper is enough, the Printer Controller calls the Print() function:
 - i. Once the printing is finished, the Printer Controller sends a message to the Print Controller.
 - ii. The Print Controller updates the printing status and saves log, then it sends a “Finished Printing” notification to the user.

- b. If the number of paper is not enough, the Printer Controller will push the request to the wait-queue of the Print Controller:
 - i. The Print Controller sends a “Printer is needed refilled paper” notification to SPSO.
 - ii. SPSO refills paper and clicks the “Complete” button on the printer’s dashboard.
 - iii. The printer sends message “Complete refill” to the Printer Controller, which forwards the message to the Print Controller.
 - iv. The Print Controller retrieves the first request from the “wait-queue” and pushes it to the head of Printer Controller’s print-queue. (Back to step 1).



2.3 Class Diagram

- **Class User**

- *Main feature:* Store the username and password of user
- *Attribute:*

- username: string
 - password: string
- **Class Student**
 - *Main feature:* Represent the student
 - *Attribute:*
 - page: int
- **Class SPSO**
 - *Main feature:* Represent the SPSO
- **Class HomeUI**
 - *Main feature:* Display the system home page for user to log in
 - *Method:*
 - *openUI()*: Open the interface for user to log in
 - *selectStudentButton(StudentUI)*: Open the interface for student to log in
 - *selectSPSOButton(SPSOUI)*: Open the interface for SPSO to log in
- **Class HomeController**
 - *Main feature:* Manage the HomeUI
 - *Attribute:*
 - User_pointer: User
 - studentUI: StudentUI
 - SPSO_UI: SPSOUI
 - *Method:*
 - *authenticate(HCMUT_SSO)*: Authenticates a user with a particular UI
 - *open_Student_UI()*: Opens the student UI
 - *open_SPSO_UI()*: Opens the SPSO UI
- **Class HomeUI_singleton**

- *Main feature:* Holds a single instance of HomeUI for singleton pattern
- *Attribute:*
 - HomeUI_instance: HomeUI
- *Method:*
 - *getInstance():* Returns the singleton instance of HomeUI

- **Class StudentUI**

- *Main feature:* Display the initial user interface specifically for student
- *Attribute:*
 - stuUIcontroller: StudentUIController
- *Method:*
 - *openUI():* initialize and display the StudentUI
 - *selectPrint():* navigate to the Upload and Specify UI when user clicks on the button
 - *selectLog():* navigate to the StudentLog UI when user clicks on the button
 - *selectMaintenance():* navigate to the Maintenance UI when user clicks on the button
 - *selectGuideline():* navigate to the Guideline UI when user clicks on the button
 - *selectContact():* navigate to the Contact UI when user clicks on the button
 - *selectHomebutton():* return to Home page
 - *selectLogout_button():* logout

- **Class StudentUIController**

- *Main feature:* Manage the StudentUI
- *Attribute:*
 - request: Request
 - upload_SpecifyPropUI: Upload_SpecifyPropUI

- choose_PrinterUI: ChoosePrinterUI
- student_LogUI: StudentLogUI
- guideline_UI: GuidelineUI
- contact_UI: ContactUI
- *Method:*
 - *open_Choose_PrinterUI()*: initialize and display the Choose_Printer UI
 - *open_Upload_SpecifyPropUI()*: initialize and display the SpecifyProp UI
 - *open_LogUI()*: initialize and display the Student_Log UI
 - *open_Maintenance()*: initialize and display the Maintenance UI
 - *open_GuidelineUI()*: initialize and display the Guideline UI
 - *sendRequest2Print(printer_id)*: send the printing request to the Controller
 - *sendNotiSPSO(printer_id)*: send a notification to the SPSO when the printer runs out of paper
 - *sendNotiStudent(student_id)*: send a notification to the student when the document is successfully printed
 - *packageRequest()*: package a request (including all properties)
- **Class Request**
 - *Main feature*: represent the printing request, storing all detail of request
 - *Attribute*:
 - requestID: int
 - stuID: int
 - filePath: string
 - property: Property
 - printerID: int
- **Class Upload_SpecifyPropUI**

- *Main feature*: Display a user interface for student to upload file and specify properties
- *Attributes*:
 - uploadController: UploadController
 - spController: SpecifyPropController
- *Methods*:
 - *openUI()*: initialize and display the interface to user
 - *uploadFile()*: display a area for student to upload a file and store it
 - *inputSpecifyProp()*: store the properties that student inputs into the interface
 - *checkValid_button()*: check the validity of the file and its properties when student clicks on the button
 - *selectHomebutton()*: return to Home page
 - *selectLogout_button()*: logout

- **Class SpecifyPropController**

- *Main feature*: Manage the property specifying process of student
- *Attribute*:
 - property: Property
- *Methods*:
 - *Check_Valid()*: Check the validity of properties
 - *displayPageOption()*: Display options “Buy more paper” or “Adjust page” for student to choose when page balance is not enough.
 - *displayError()*: display the error message when the properties is invalid
 - *adjustPage()*: clear all the properties in the interface to specify again when student clicks on the button

- *buyPage()*: navigate to the BK Pay when student clicks on the button
- *Printerbutton()*: navigate to the Choose Printer UI when student clicks on the button.

- **Class UploadController**

- *Main feature*: Manage the file uploading process of student
- *Attributes*:
 - *uploadStatus*: bool
 - *file_name*: string
- *Methods*:
 - *select_file()*: allow student to select a file from their local device for upload. When student clicks on the button, it opens a file dialog box where the student can browse their folders and choose the file they want to upload.
 - *drag_file()*: allow student to upload a file by dragging it into a designated area in the UI.
 - *checkFileSize(file_name)*: Check the validity of file size
 - *checkFiletype(file_name)*: Check the validity of file type
 - *notiStudent()*: display the error message when the file is invalid

- **Class ChoosePrinterUI**

- *Methods*: Display a interface for student to choose printer for printing
- *Attributes*:
 - *cpController*: ChoosePrinterController
- *Methods*:
 - *openUI()*: initialize and display the interface to user
 - *displayPrinterList()*: display the list of printers

- *selectPrinter()*: display the details of printer when student clicks on a printer's view button in the interface
- *selectHomebutton()*: return to Home page
- *selectLogout_button()*: logout
- *searchPrinter()*: update the list of printer when student inputs the detail into the Search box

- **Class ChoosePrinterController**

- *Main feature*: Manage the printer choosing process of student
- *Attributes*:
 - printerCache: list<Printer>
- *Methods*:
 - *updateCache()*: update the latest list of printer in the interface
 - *checkAvailable(ID)*: Check if the printer is available when student clicks on the printer
 - *noticeStudent()*: Display a message "Printer may take some time" and options "Choose again" or "Continue" for student
 - *confirm_Message()*: display a confirmation message for student if the printer is available or student selects "Continue".

- **Class Property**

- *Main feature*: store all details of document that student specifies
- *Attributes*:
 - paperSize: enum
 - paperRange: list<int>
 - sideOption: bool
 - numCopies: int

- **Class GuidelineUI**

- *Main feature*: Display a interface for student to read the guideline of system

- *Methods:*
 - *openUI()*: initialize and display the interface to user
- **Class ContactUI**
 - *Main feature*: Display a interface for student to read the contact detail of system
 - *Methods:*
 - *openUI()*: initialize and display the interface to user
- **Class MantainanceUI**
 - *Main feature*: Display a interface for student to read the maintenance detail of system
 - *Methods:*
 - *openUI()*: initialize and display the interface to user
 - *Change_Page()*: display the second page of the interface
- **Class StudentLogUI**
 - *Main feature*: Display a interface for student to view the details of printing logs
 - *Attribute:*
 - viewStuLogController: ViewStudentLogController
 - *Methods:*
 - *openUI()*: initialize and display the interface to user
 - *displayListStuLog()*: display the list of printing logs in the interface
 - *selectStuLog()*: display the log when student clicks on a log.
 - *selectViewDetail(stuID)*: display the details of log when student clicks on a log's "View" button.
 - *search_Log()*: update the list of logs when student inputs the detail into the "Search" box
 - *selectHomebutton()*: return to Home page

- *selectLogout_button()*: logout
- **Class StudentLogController**
 - *Main feature*: Manage the the StudentLogUI
 - *Methods*:
 - *getLogFromDB()*: Get the latest detail of logs from the database
- **Class SPSOUI**
 - *Main feature*: Display main interface for SPSO.
 - *Attribute*:
 - *spsoController*: SPSOUI_Controller
 - *Method*:
 - *openUI()*: initialize and display the interface to user
 - *select_PrinterList()*: switch to Printer Status UI when user clicks on the button
 - *select_AnnualReport()*: switch to Annual Report UI when user clicks on the button
 - *selectHomebutton()*: return to Home page
 - *selectLogout_button()*: logout
- **Class SPSOUI_Controller**
 - *Main feature*: control opening next interface of SPSOUI
 - *Attribute*:
 - *printer_statusUI*: PrinterStatusUI
 - *report_UI*: ReportUI
 - *configUI*: ManageSysConfigUI
 - *Method*:
 - *open_printerStatusUI()*: open the UI of the attribute printer_statusUI
 - *open_reportUI()*: open the UI of the attribute report_UI

- *open_configUI()*: open the UI of the attribute configUI

- **Class ManageSysConfigUI**

- *Main feature*: initialize and display the Manage configuration interface to user
- *Attribute*:
 - mscController: ManageSysConfigController
- *Method*:
 - *displayCurrentSysConfig()*: display the UI
 - *selectChangeDefaultPagesButton()*: change default pages when clicks on button
 - *selectChangeDateGivePageButton()*: change the date give page when clicks on button
 - *selectChangeFilePermittedButton()*: change file permitted when clicks on button
 - *selectSaveChangeButton()*: save all changes when clicks on button

- **Class ManageSysConfigController**

- *Main feature*: control the changes when any button is clicked
- *Method*:
 - *getCurrentSysConfigFromDB()*: get the newest configuration of system from database
 - *updateDefaultPages2DB()*: update change of default pages to database
 - *updateDateGivePage2DB()*: update change of date give page to database
 - *updateFilePermitted2DB()*: update change of file permitted to database

- **Class PrinterStatusUI**

- *Main feature*: initialize and display the interface of printer status to user
- *Attribute*:
 - psController: PrinterStatusController
- *Method*:
 - *selectHomeButton()*: return to Home page
 - *selectLogout_button()*: logout
 - *searchPrinter()*: search printer based on the id typed in the box
 - *selectAddPrinter()*: open the add-printer window when clicks on the button
 - *selectSetting(Printer_id)*: select the setting system of 1 printer
- **Class PrinterStatusController**
 - *Main feature*: control opening process of PrinterStatusUI
 - *Method*:
 - *getPrinterListFromDB()*: get printer list to display
 - *addNewPrinter()*: add new printer
 - *change_PrinterSetting(id)*: change printer setting when clicks on the setting button
- **Class ReportUI**
 - *Main feature*: display the report interface
 - *Attribute*:
 - reportController: ReportController
 - *Method*:
 - *displayReport()*: display report UI
 - *selectHomeButton()*: return to Home page
- **Class ReportController**
 - *Main feature*: control opening report
 - *Method*:

- *getMonthlyReportFromDB()*: take monthly report from database
- *getAnnuallyReportFromDB()*: take annually report from database

- **Class Controller**

- *Main feature*: main control part of the system, interact between frontend and backend
- *Attribute*:
 - PrinterController_List: <PrinterController>
 - database_Controller: DatabaseController
- *Method*:
 - *receiveRequest()*: receive the request sent from StudentUIController
 - *finishPrint(Request)*: finish print process and send notification back to student
 - *check_wait_queue()*: check all wait_queue of printer lists, if there is a wait_queue that is not empty, then notice SPSO about the page lack.
 - *addprinter2DB(printer)*: when SPSO adds a printer from frontend, the controller receives information and adds it to the database.

- **Class DatabaseController**

- *Main feature*: connect and interact between application and database.
- *Method*:
 - *getPrinterList()*: get printer list from database
 - *getUser(username)*: get user information from database

- **Class PrinterController**

- *Main feature*: a controller represents the printer

- *Attribute:*

- run_queue: queue<Request>
- wait_queue: queue<Request>
- database: DatabaseController
- printer: Printer

- *Method:*

- *pushRequest(Request)*: push a new request to run queue.
- *print()*: takes the first request in the front of wait queue to execute, if it can not execute then take the request in the front of run_queue to execute and print.
- *pushWaitRequest(Request)*: push a request to wait queue.
- *finishPrint()*: notification to student
- *checkPageRemain()*: update number of paper of printer and student after printing.

- **Class Printer**

- *Main feature:* a struct for printer

- *Attribute:*

- PrinterID: int
 - status: bool
 - numPaper: int.

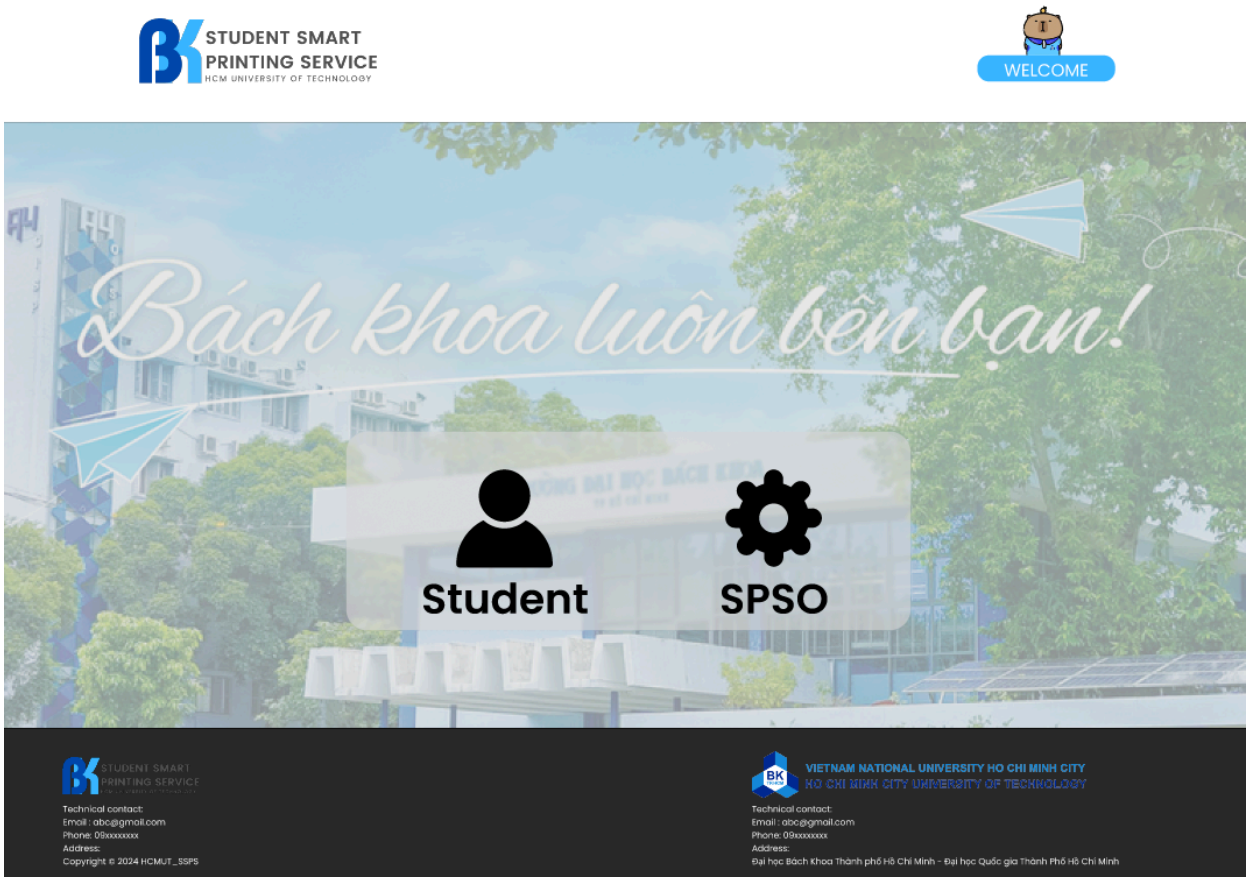
2.4 User Interface for printing module

Link for figma:

<https://www.figma.com/proto/JabtJcbGPZNKtcAclNbDSZ/CNPM---FIGMA?node-id=693-2442&node-type=canvas&t=KioOb6goZd1zzdLT-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=693%3A1074&show-proto-sidebar=1>

2.4.1 Home UI

When we move the mouse over the main screen, a login button will appear. Click the button will give two options, login as student or as SPSO like the figure shown below.



Name	Type	Description
Log in	Button	Navigate user to log in session
User	Button	Choose to log in as a user
SPSO	Button	Choose to login as SPSO

2.4.2 Login UI



Welcome!

Username

Password

ĐĂNG NHẬP

STUDENT SMART
PRINTING SERVICE

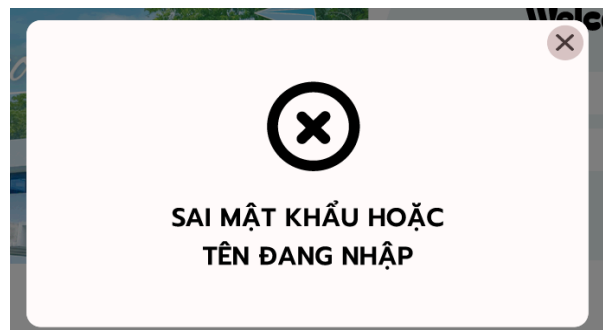
Technical contact:
Email: abc@gmail.com
Phone: 09xxxxxxx
Address:
Copyright © 2024 HCMUT_SSPS

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Technical contact:
Email: abc@gmail.com
Phone: 09xxxxxxx
Address:
Đại học Bách Khoa Thành phố Hồ Chí Minh - Đại học Quốc gia Thành Phố
Hồ Chí Minh

Name	Type	Description
Username	Input	Require user to enter username
Password	Input	Require user to enter password
Login	Button	Log in to the system

If a user enters an invalid username or password, this pop up will appear



User need to close it and re enter a valid username or password

2.4.3 Student Home Page



Maintenance Information

[View](#)

Guideline - Instruction

[View](#)

Contact - Support

[View](#)

Name	Type	Description
Home	Button	Go back to the main screen of student
Log out (label on the top right)	Button	Log out, go to the login screen
In tài liệu	Button	Navigate user to uploading and specify properties interface before print the document
Xem lịch sử	Button	Navigate user to print log, display all the printing history of user
Maintenance information	Button	Navigate user to maintenance informations
Guideline - Instruction	Button	Navigate user to guideline-instruction
Contact - Support	Button	Navigate user to the bottom of the screen, which contains contact information

2.4.4 Upload and Specify properties UI

[HOME](#)
[PRINT](#)

ABCD
Số trang: 15

Kéo thả tệp tin hoặc

Chọn tệp

.pdf, .docx, .pptx
Độ lớn tệp tin tối đa 50MB

Các tệp tin đã tải lên

Example.docx

Example.docx

Example.docx

Khổ giấy

Chọn khổ giấy

Số trang

Chọn số trang

Một mặt ☐
Hai mặt ☐

Tùy chọn lật cạnh

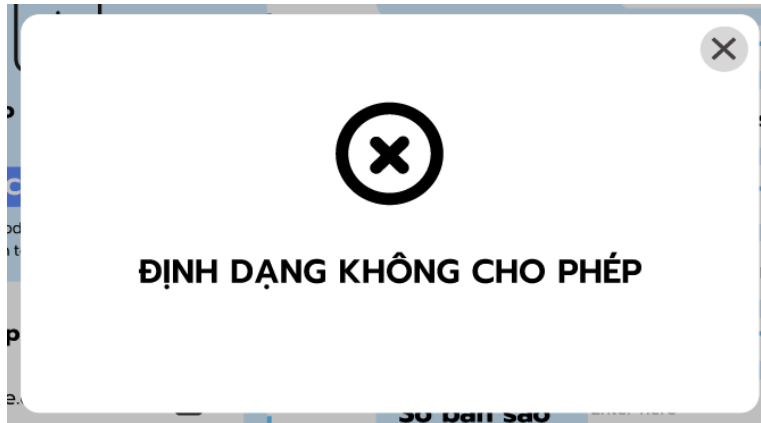
Số bản sao

Enter here

Kiểm tra

Name	Type	Description
Chọn tệp	Button	Browse files from user
Số trang	Drop-down table	Select number of pages to be printed
Khổ giấy	Drop-down table	Size of the printed document (A3, A4)
Số mặt	Button	Print on 1 side or both side of page
Số bản sao	Input from user	Number of copies
Tùy chọn lật cạnh	Drop-down table	Flip on short or long edge
Delete (Bin symbol)	Button	Delete the file
Kiểm tra	Button	In order, check the validity of the request

If the file format is invalid show a pop up “Error file type”:



If the user's page balance is not enough, show a pop up “not enough page balance”:



If user choose “Buy more paper”, show this pop up:



If the request is valid, show a pop up “Choose printer” for the user:



2.4.5 Choose printer UI


[HOME](#)
[PRINT](#)



ABCD

Số trang: 15

ID MÁY IN	VỊ TRÍ	TÌNH TRẠNG	CHỌN
PRINTER#1	402A4	Bật	☐
PRINTER#2	402A4	Bật	☐
PRINTER#3	402A4	Bật	☐
PRINTER#4	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐
PRINTER#...	402A4	Bật	☐

Kiểm tra

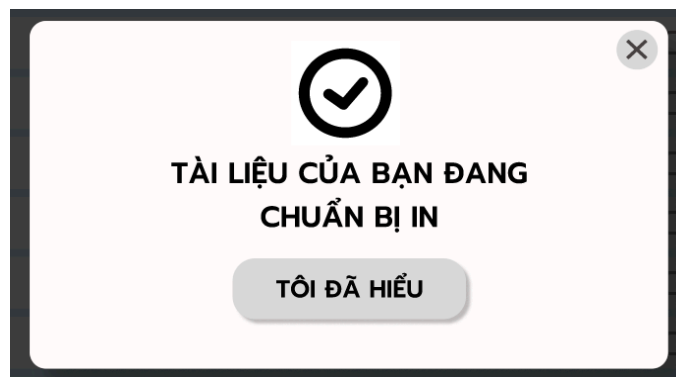
Name	Type	Description
Nhập ID máy in	Input	Input a printer id to filter out specific printer
Chọn	Radio	Choose a printer

Kiểm tra	Button	Check if the chosen printer is available
----------	--------	--

If the selected printer is busy or temporarily out of paper, this pop up will be shown:



If user choose “Chọn lại” button, it will return to Choose printer UI, otherwise, choose “Tiếp tục”, this pop up will be displayed:



If the user chooses the “x” button, the print request will be discharged.

2.4.6 Student view log UI


HOME PRINT



ABCD

Số trang: 15



Từ:

Nhập ngày

Đến:

Nhập ngày

Q

Nhập tên file

STT	NGÀY	TÊN FILE	NƠI NHẬN	TÌNH TRẠNG	CHI TIẾT
20	20/11/2024	File name	402A4	Đang in	
19	20/11/2024	File name	402A4	Đang in	
18	19/11/2024	File name	402A4	Chưa nhận	
17	19/11/2024	File name	402A4	Chưa nhận	
16	20/10/2024	File name	402A4	Đã nhận	
15	10/10/2024	File name	402A4	Đã nhận	
14	1/10/2024	File name	402A4	Đã nhận	
13	3/9/2024	File name	402A4	Đã nhận	
12	3/7/2024	File name	402A4	Đã nhận	

Name	Type	Description
Từ (ngày)-Đến (ngày)	Input	Input to filter out the history of all documents printed in an interval of time
Nhập tên file	Input	Input a name to filter out the history of all documents printed with the most same name
Chi tiết	Button	View a single printing request in detail

The detail include the following field:

×

• Tên file:

Filename.pdf

• Ngày bắt đầu in:

20/11/2024

• Ngày nhận:

22/11/2024

• Nơi nhận:

402A4-CS1

• ID máy in:

Printer#1

• Khổ giấy:

A4

• Số trang:

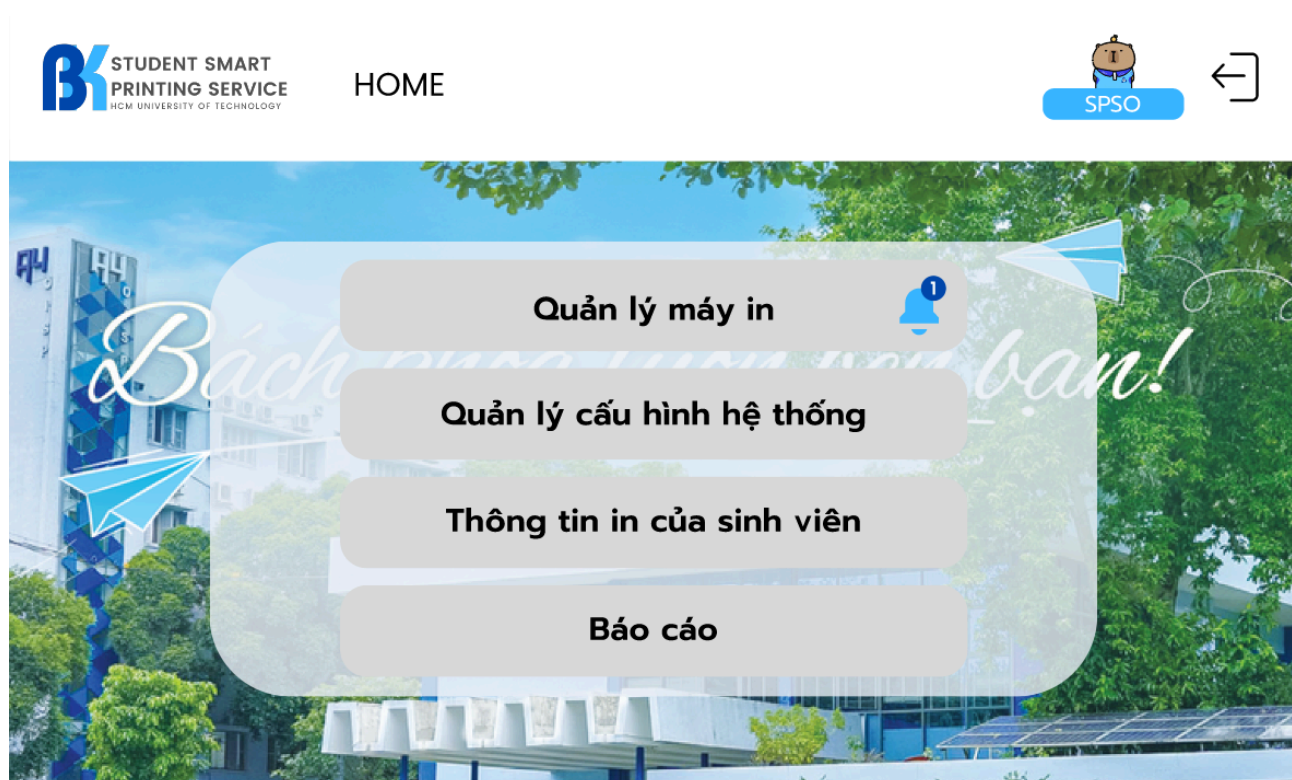
Tất cả trang

• Hai mặt

• Số bản sao:

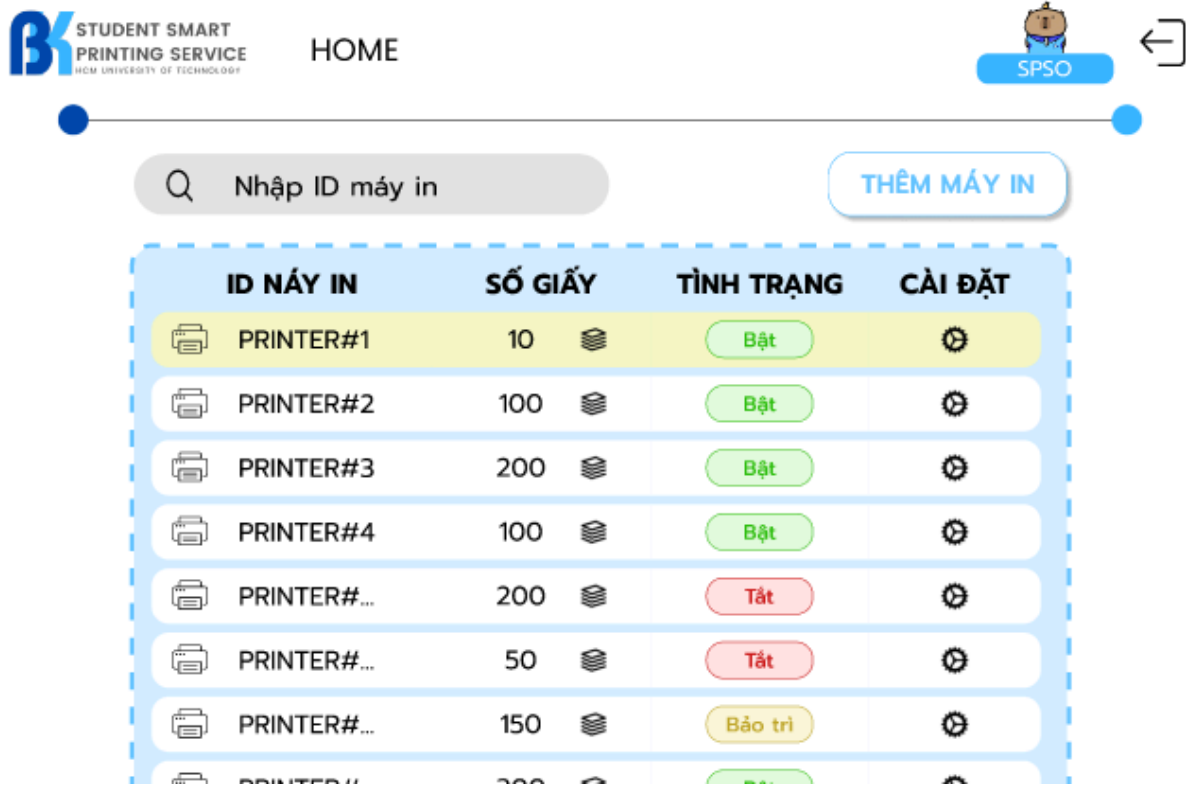
1

2.4.7 SPSO Home Page



Name	Type	Description
Home	Button	Go back to the main screen of SPSO
Log out (label on the top right)	Button	Log out, go to the login screen
Quản lý máy in	Button	Go to the manage printer interface
Thông báo	Icon	Notify there are some printers that have printing request but out of paper
Quản lý cấu hình hệ thống	Button	Go to manage configuration interface
Thông tin in của sinh viên	Button	Go to view student printing log interface
Báo cáo	Button	Go to view monthly and annual report

2.4.8 Manage Printer UI



Name	Type	Description
Home	Button	Go back to the main screen of SPSO
Thêm máy in	Button	Add new printer and it's status
Nhập ID máy in	Input	Search printer by its ID
Xấp giấy (in Số giấy column)	Button	Refill the paper, when a printer has not enough paper to print, its row will change to yellow color
Cài đặt	Button	Config the status of a printer

When click a pop up is showed and SPSO can choose the printer status as following

• ID máy in

Printer#1

• Tên máy in:

ABC

• Địa chỉ IP:

111.111.1.111

• Kết nối:

Kết nối

• Tình trạng

Bật

Điều chỉnh trạng thái

Tình trạng

Bật

Tắt

CON/DIS

Kết nối

Không kết nối

Hủy

LƯU

Name	Type	Description
Tình trạng	Drop down table	Turn on or turn off the printer status
CON/DIS	Drop down table	Connect or disconnect the printer
Hủy	Button	Discard the changes have been made
Lru	Button	Save the changes

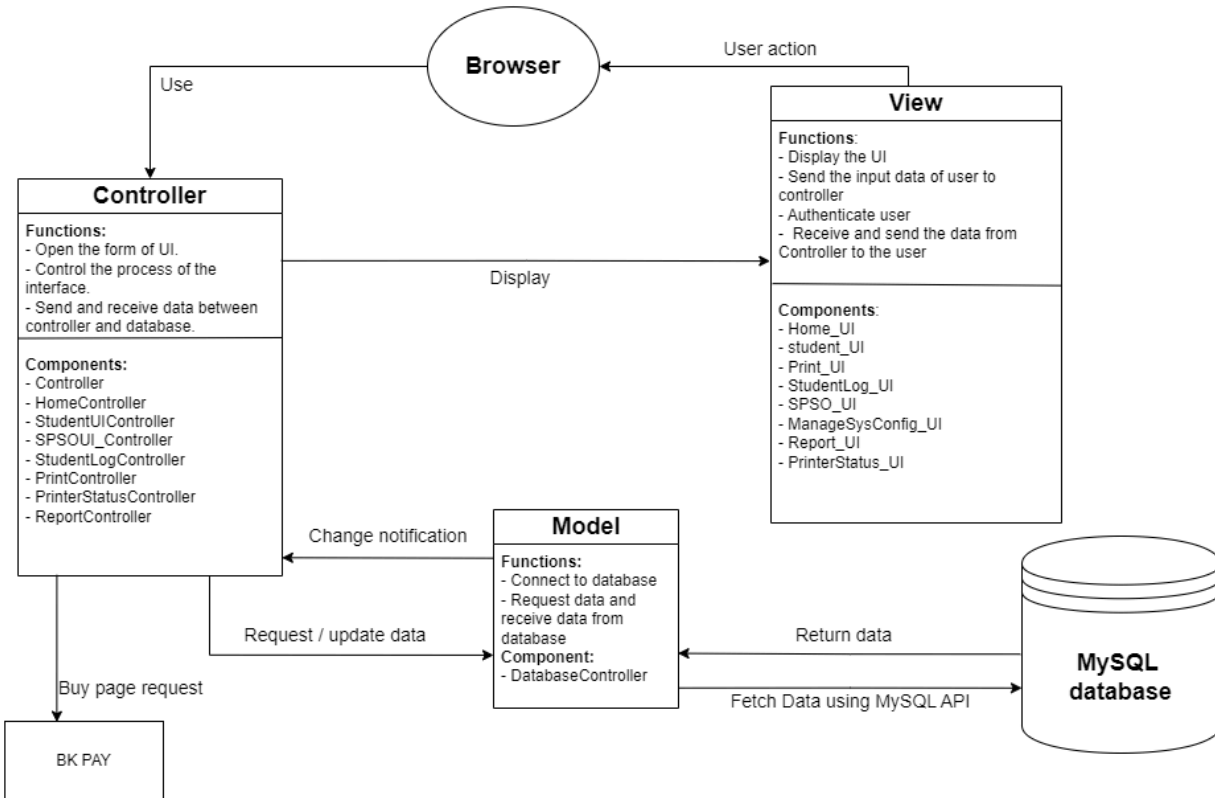
2.4.9 Manage Configuration UI

Name	Type	Description
Home	Button	Go back to the main screen of student
Log out (label on the top right)	Button	Log out, go to the login screen
Số giấy mặc định	Input	Enter the default page balance for student to receive after a period of time (applied to all students in the system)
Ngày cấp giấy	Input	Enter the date such that the system will reset the page balance for all students (in dd/mm/yyyy format)
Định dạng cho phép	Button	Choose the file format that will be accepted by the SPSS (applied for all printers in the system)
Lưu thành đổi	Button	Check for validity of all inputs (date or page balance) then apply the config for the whole system

3. Task 3: Architecture design

3.1 Layer architecture

3.1.1 MVC architecture



MVC pattern to design the HCMUT-SSPS system

To design the HCMUT-SSPS system, we decided to use the MVC (Model-View-Controller) architectural model. We chose this architectural model for several reasons:

- Each feature in the SSPS system usually requires multiple interfaces (Views) and involves various data-related operations. For example, the main document printing feature requires displaying information about uploading files, specifying printing properties, and choosing printers across different interfaces. Each of these interfaces often needs to handle actions like viewing, editing, and updating the corresponding data. Using the MVC architecture clearly separates the main components: Model, View, and Controller. The

Model is responsible for data, the View handles displaying information to users, and the Controller manages data-processing logic. This separation makes the codebase clear and easier to maintain.

- The separation between Model, View, and Controller also makes it easier to expand the system. Any new requirement related to data interaction can be added to the system without affecting the user interface and vice versa. This enables convenient and independent integration and expansion of new features.

However, MVC architecture also comes with the complexity of ensuring effective interaction between the Model, View, and Controller. To ensure these components work together cohesively, technical expertise and tight project management are required to maintain the stability and maintainability of the system.

Here is a description of the MVC architecture that we use to design HCMUT-SSPS:

- **Browser:** This is where users interact with the system, entering and retrieving information through the interfaces.
- **View:** This component is responsible for managing the user, including setting up the user interface elements, notifications, and pop-up messages, forms across the system. It also responds to user actions taken on the browser, such as entering or modifying data and requesting data views. These actions are managed by the View and forwarded to the Controller through the Browser as requests. After processing, the results are sent back to the View, which updates the display based on user actions.
- **Controller:** This is the system's main layer, serving as the central processing component, the Controller manages communication between the View and the Model. Its key functions include:
 - **Request processing:** The Controller receives requests from the View, retrieves necessary data from the Model, and processes the request using that data. Once processing is complete, the Controller sends data

back to the View to update the display. This function makes the Controller the core of the system's processing.

- **Data validation:** The Controller checks data validity for actions requested by the View, whether for data retrieval or user updates intended for the database.
- **Data updates:** In addition to retrieving data from the Model, the Controller also updates new data in the database. When user actions or system processes require changes to stored data, the Controller communicates with the Model to execute these updates, ensuring the database remains current and accurately reflects user or application changes.
- **Model:** This part manages data, including basic functions related to data processing. In the HCMUT-SSPS system, the Model serves as the primary communication link with our MySQL database, which stores the system's data. It receives data updates from the Controller and applies them to the database, as well as retrieves requested data.

3.1.2 Presentation strategy

In the last task, we already prepared a figma design of our web page. In order to turn the figma design into a working app, we need to study the design carefully. Look at each part and components, paying to the details like sizes, colors, and font. Then, we need to set up a front-end framework (we choose ...) after considering to meet the requirements of the assignment and also to match the style of our design. Next, we will define global styles to match Figma's colors, fonts, and spacing, so everything stays consistent. We will begin to build small, reusable components, like buttons or drop tables, and then combine them into bigger parts of the page. Because the design is only for laptops, we don't need to adjust the layout based on screen size. We will add interactivity such as smooth transitions or animations. Finally, the work will be tested by comparing it with the Figma design to make sure it looks the same.

3.1.3 Data storage approach

To handle the problem of data storage, we decided to use MySQL relational database management system. MySQL is an open-source relational database that uses structured query language to manage and organize data. It is widely used for its speed, reliability and ease of use.

Features of MySQL:

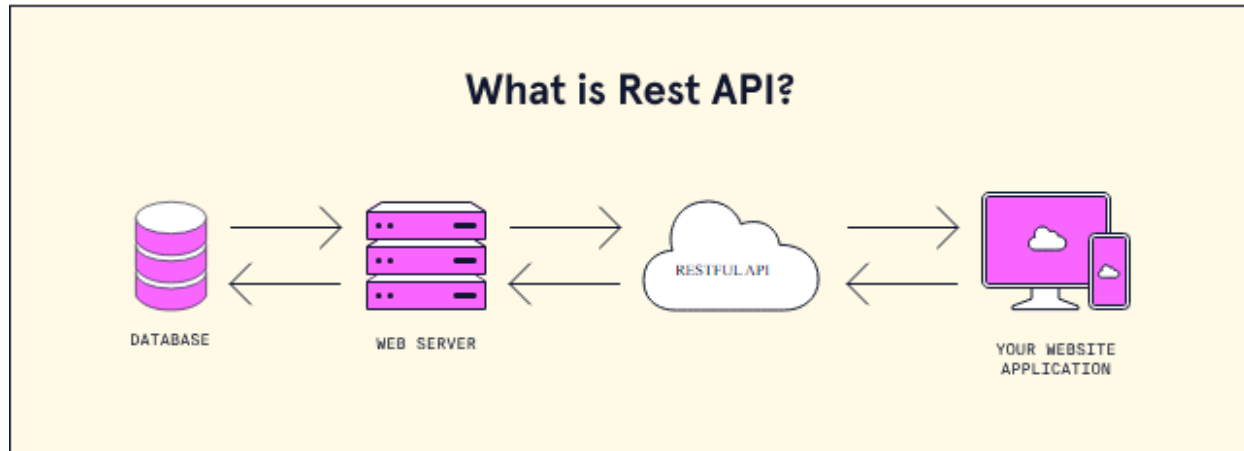
- **High performance:** MySQL is optimized for fast query processing, making it suitable for high-demand applications like e-commerce web-based applications.
- **Scalability:** MySQL supports large databases with up to millions of records and can scale horizontally by adding more servers.
- **Secure data storage:** MySQL provides security services such as data encryption, authentication as well as SSL (Secure Sockets Layer) support for secure connections.
- **Backup and recovery options:** MySQL includes backup utilities and point-in-time recovery options.

HCMUT Smart Printing Service Database has several tables as the strong entities of the system. This includes:

- **Printer:** this table stores information about all printers such as location, amount of papers and status.
- **Report:** all of the reports (include annually, monthly and daily) store here.
- **Request:** Each student when using the system to print a document needs to send a specific request to the controller. This table stores all of the requests in case the system needs to generate a report.
- **User:** all of the accounts information will be stored in this table. We decided to separate users into two types: SPSO and student.

3.1.4 API management

To connect and exchange data between programs, we decided to use REST API (Representational State Transfer). REST is an architecture style that uses HTTP requests to communicate with a server to perform operations on resources such as GET, POST, PUT, DELETE.

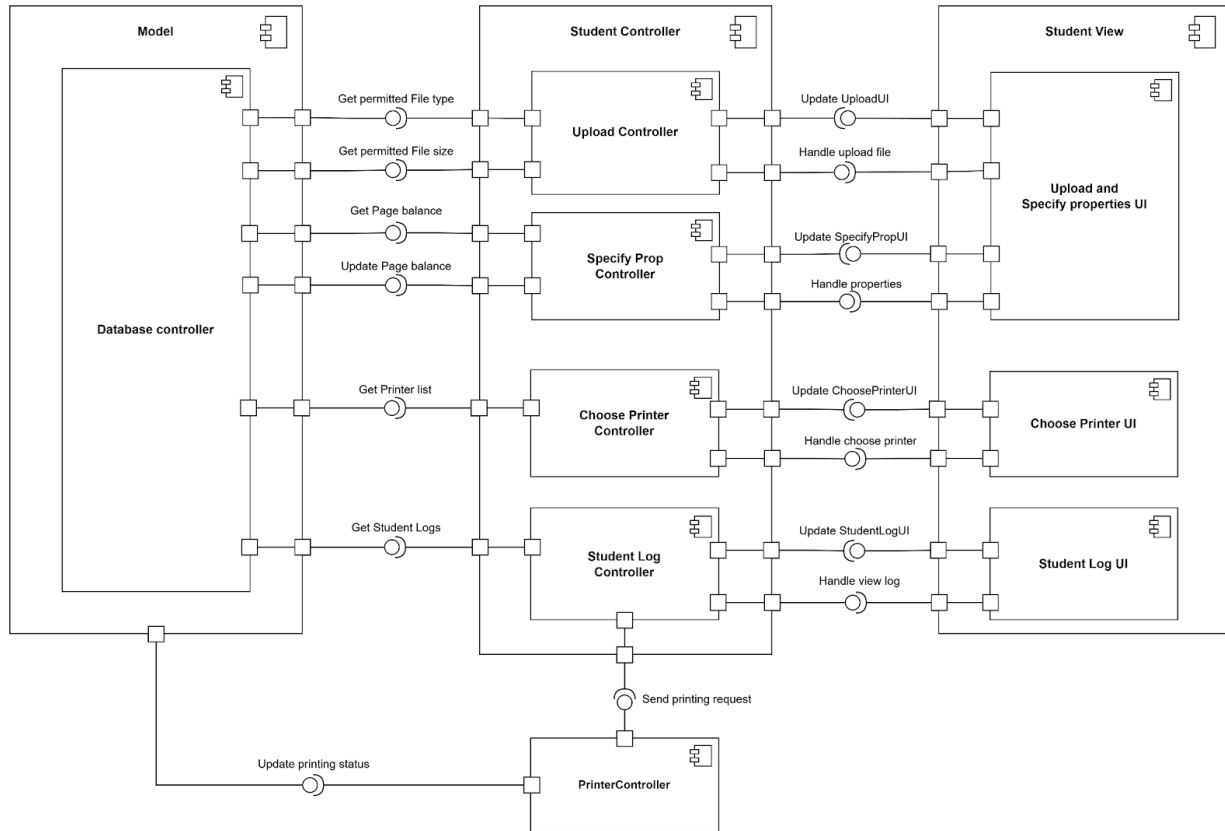


In this architecture, a request is sent from client to server in the form of a web URL as HTTP GET, POST, PUT or DELETE request, and then the server will send back a response in JSON, HTML, XML or Image.

- GET: method is used to read a representation of a resource.
- POST: method is used to request the server to accept the resource, then it is most often utilized to create or update resources.
- PUT: is also a method that is used for sending resources to server. The difference between POST and PUT is that PUT can allow client to position the resource in the right known place while POST will let the server do that.
- DELETE: delete a resource.

Because of the simplicity, scalability and flexibility, REST API has become a standard in modern web development for creating applications that communicate over web service.

3.2 Component diagram



Component diagram of printing module

3.2.1 Student View Component

The **Student View Component** contains classes responsible for the user interface that enables students to request document printing. It includes:

- **Upload and Specify properties UI**
 - Display the user interface that allows students to upload and specify the printing properties of a document.
 - It provides an interface to receive *updates of view* from the **Upload Controller** and *handle the upload process* such as validating file types and file sizes to ensure they meet permitted requirements.
 - It also provides an interface to *receive updates of view* from the **Specify Property Controller** and *handle the input properties* from students such as

validating that the student's page balance is sufficient for printing the uploaded document.

- **Choose Printer UI**

- Offers the user interface for students to select a printer for the uploaded document
- It provides an interface to *receive updates of view* from the **Choose Printer Controller** and *handle the choosing process* such as checking to ensure the selected printer is available (e.g., it is "on") and has sufficient resources to complete the printing job.

- **Student Log UI:**

- Offers the user interface for students to view their printing log
- It provides an interface to *receive updates of view* from the **Student Log Controller** including updating the status of printing process (in printing, received or unreceived), the date that student received the printed document.

3.2.2 Student Controller Component

The **Student Controller Component** contains classes responsible for managing tasks when students request document printing. It includes:

- **Upload Controller:**

- It requires an interface for **Upload and Specify properties UI** to make *updates on the view*
- It provides an interface for **Upload and Specify properties UI** to *handle the the upload process* when student drag and drop or click "Browse" button
- It requires an interface provided by the **Database Controller** to interact with the database and retrieve the data about permitted file types and file size.

- **Specify Property Controller:**

- It requires an interface for **Upload and Specify properties UI** to make *updates on the view*

- It provides an interface for *Upload and Specify properties UI* to handle the input printing properties process
- It requires an interface provided by the *Database Controller* to interact with the database and retrieve the data about the student's page balance as well as update the latest page balance value.
- **Choose Printer Controller:**
 - It requires an interface for *Choose Printer UI* to make updates on the view
 - It provides an interface for *Choose Printer UI* to handle the choosing printer process when students click the “Check valid” button
 - It requires an interface provided by the *Database Controller* to interact with the database and retrieve the data about the printers.
- **Student Log Controller:**
 - It requires an interface for *Student Log UI* to make updates on the view
 - It provides an interface for *Student Log UI* to handle the viewing process when students click the “View” icon
 - It requires an interface provided by the *Database Controller* to interact with the database and retrieve the data about the student's printing log.

The Student Controller also requires an interface provided by the *Printer Controller* to send print requests.

3.2.3 Printer Controller Component

- It provides an interface that allows the *Student Controller* to submit printing requests.
- It requires an interface provided by the *Database Controller* to update the print job status.

3.2.4 Database Controller Component

- It provides an interface that the *Upload Controller* can call to request the data about permitted file type and file size.

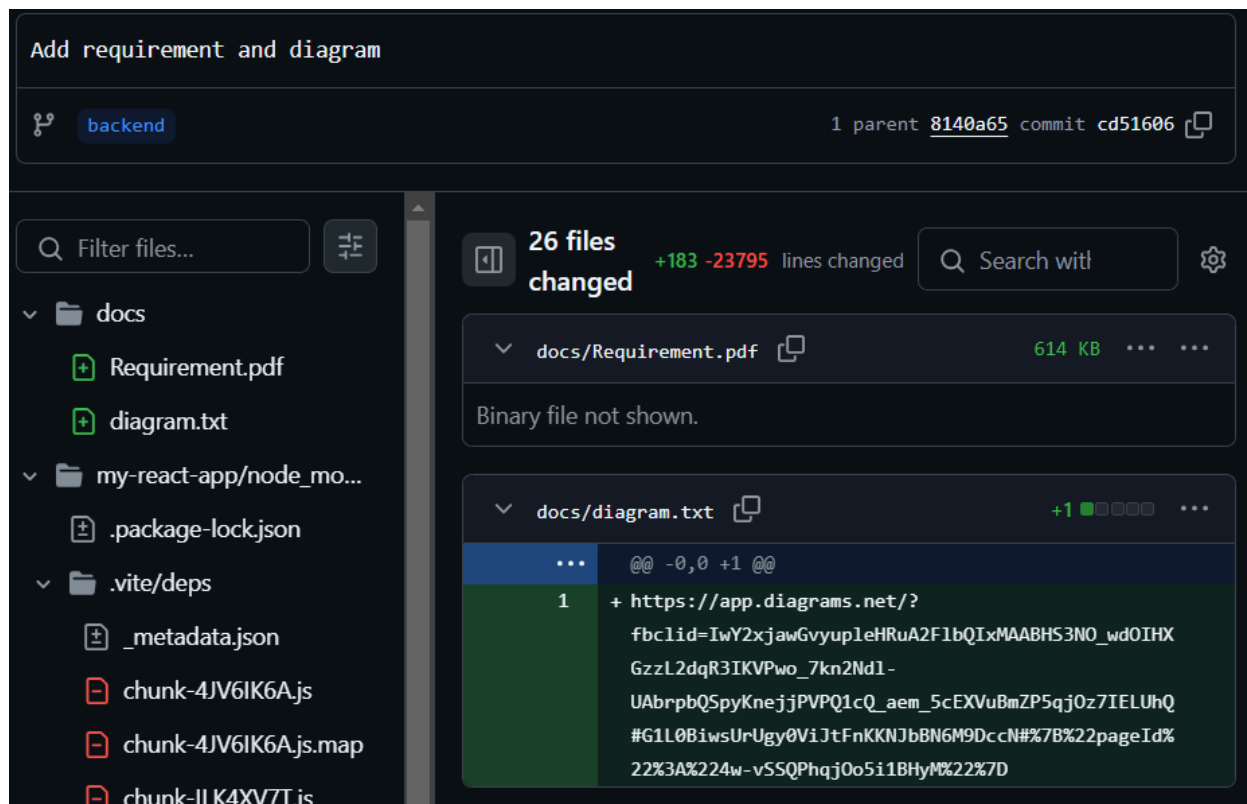
- It provides an interface that the *Specify Property Controller* can call to request the data about the student's page balance as well as update the latest page balance value.
- It provides an interface that the *Choose Printer Controller* can call to request the data about the printers.
- It provides an interface that the *Printer Controller* can update the printing status of students' documents.

4. Task 4: Implementation – Sprint 1

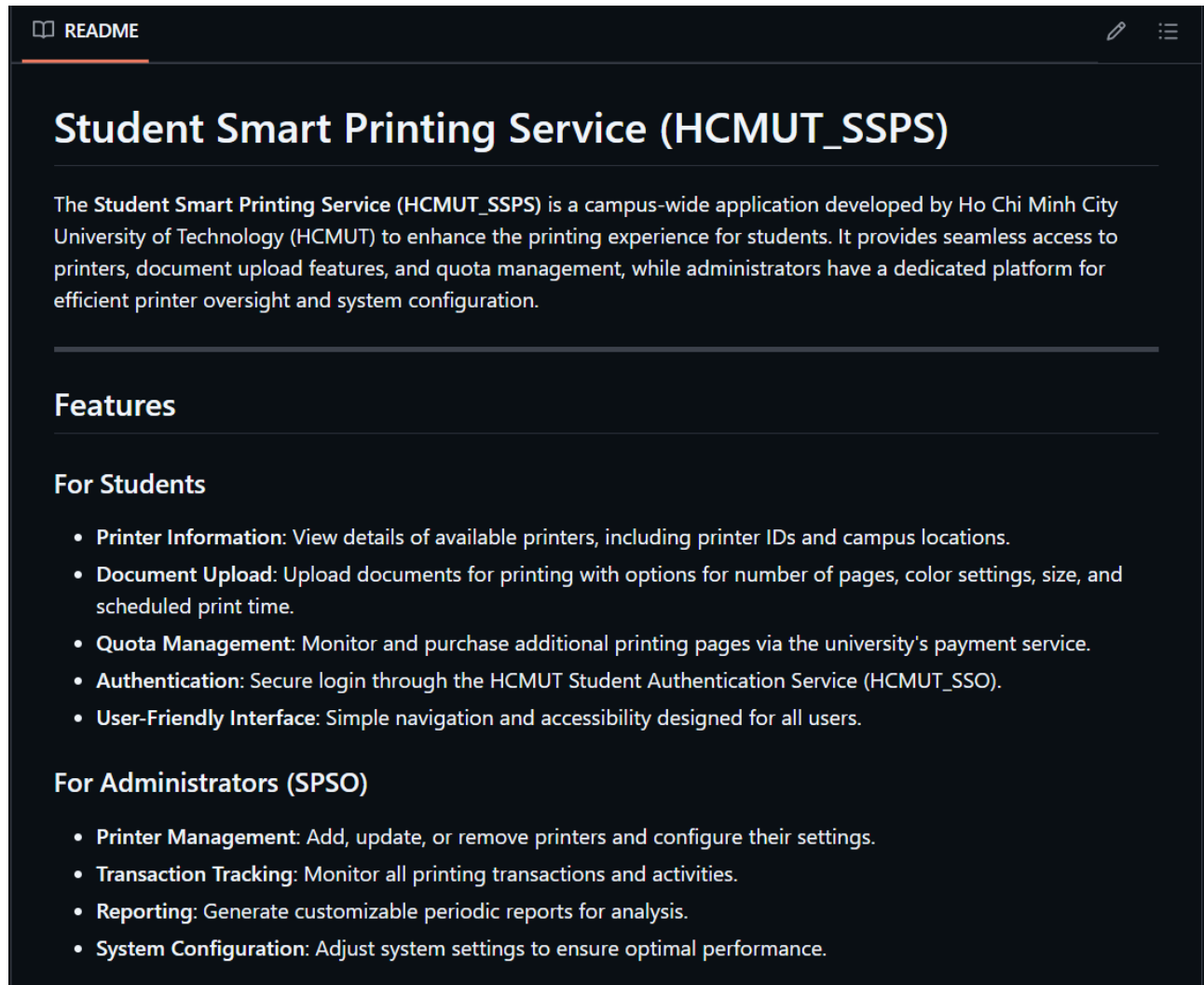
4.1 Version control

Link Github: https://github.com/PhiuTheWind/CNPM_Project

4.2 Commit and Change log



Add requirement and diagrams for System modeling and Architecture design

*README file*

4.3 Usability Testing

The members involved in the development of the SSPS printing service system conducted a usability testing session using the implemented Figma prototype available at this link: <https://shorturl.at/CdAE9>

The testing session was held with the participation of all members. During the session, the team recorded solutions, task completion rates, comments, overall evaluations, questions, and feedback from all participants.

4.3.1 Recruit participants/ testers

We use google form as tool to collect rating and feedbacks from tester

LINK: <https://forms.gle/9v6EcAxhkckXi2b87>

4.3.2 Define tasks

We outlined several tasks based on the important use-cases of the system as follows:

	Use-case	Task description
1	Home Page and Log in page	Log in to the system using valid credentials (username and password). If incorrect credentials are provided, observe the error messages displayed
2	Upload Document	Upload a file to the system. Check whether the system accepts the file and displays a confirmation after successful upload.
3	Specify properties	Specify document properties such as paper size, number, and orientation. Ensure all selected properties are displayed correctly
4	Choose printer	Select an available printer from the list. Confirm that the system updates the chosen printer and provides relevant information about the printer.
5	Student View log	Display a list of printing log of student from the latest to the oldest, choose view detail to see the log detail
6	Student printing summary	Choose the avatar to access the summary report, it provides the summary of the number of printed pages for each page size,...
7	Manage printer	Add, remove, or update printers in the system. Verify that all changes reflect correctly in the printer management list.
8	Manage configuration	Modify printer's configuration. Ensure the configurations are saved and applied accurately to the printer.
9	SPSO View student log	Review the detailed printing log of all students or one student in a period of time
10	SPSO View report	Review the monthly or annually report

4.3.3 Define test strategy

Based on the feature of the project and constraints in budget and time, our team decided to conduct testing in the following approach:

- **Remote Unmoderated Testing:** In this method, test users receive a link to the system along with an evaluation form. They interact with the application independently and complete the evaluation without direct supervision from the team.

This decision was made for the following reasons:

- **Broader Reach and Scalability:** By sending the evaluation link to multiple users, we can involve a larger pool of participants across different locations. This approach increases the diversity of feedback and ensures that more opinions are considered.
- **Cost-Effective:** This strategy eliminates the need for arranging meetings or scheduling specific sessions with each user, significantly reducing both time and logistical expenses.
- **User Convenience:** Test participants can evaluate the application at their own convenience, leading to a more relaxed and unbiased interaction with the system.

Additionally, we will use a qualitative approach. This strategy focuses on gathering detailed information and insights during the testing process, including how each user interacts with the product or service. The reason for choosing this method is that our team prioritizes user behavior and experience, making this approach more suitable for their product.

4.3.4 Conduct the test

For preparation, the team created a detailed evaluation form with clear instructions and a list of tasks for users to perform. The form included a 5-point scale, ranging from "Strongly Disagree" to "Strongly Agree," for rating criteria such as usability, design, functionality, and overall experience. Open-ended questions were added to gather qualitative insights about user challenges and suggestions. Then a unique link to the application and the evaluation form was generated and distributed to test participants via email and messaging platforms. Participants were given a deadline to complete their tasks and submit the forms. Participants were instructed to carefully read the task scenarios provided in the evaluation form and interact with the system to complete the outlined tasks. Additionally, participants were also encouraged to share additional feedback or issues they encountered during the testing process.

We use the 5-point scale, ranging from "Strongly Disagree" to "Strongly Agree." based on the following criteria:

- **Usability:** How easy and intuitive it is to use the interface.

- **Interaction flexibility:** How adaptable and versatile the interface is during use.
- **Information location accuracy:** How well users can predict where to find specific information in the interface.
- **Learnability:** How easy it is for users to learn to navigate and operate the interface.
- **Support and assistance:** How easily users can access help or find the information they need.
- **Layout design:** How well-organized and visually structured the interface is.
- **Visual appeal:** How attractive and engaging the interface is for users, encouraging exploration.

4.3.5 Evaluation results, feedback

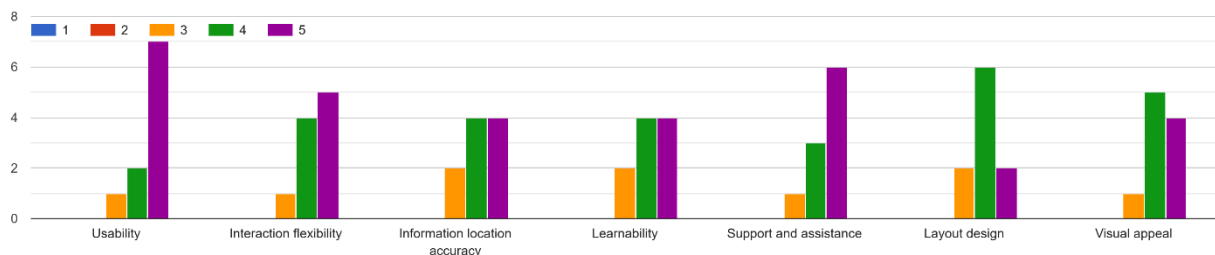
a. Evaluation results

The rating scale from 0-5 typically includes the following levels:

1. Strongly Disagree
2. Disagree
3. Neutral
4. Agree
5. Strongly Agree

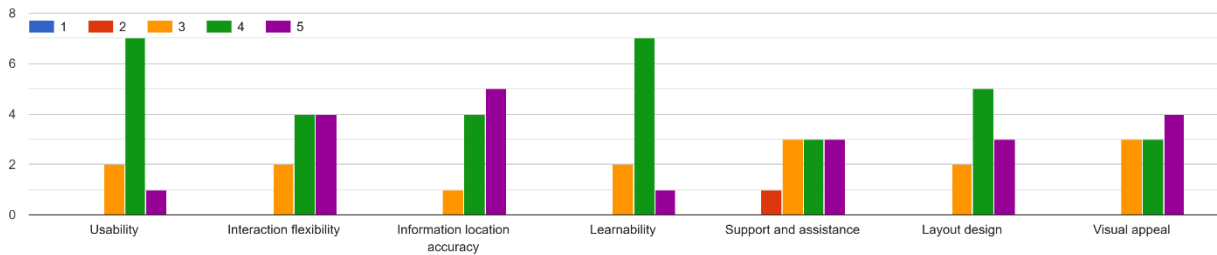
Home Page and Log in page:

Home Page and Login Page



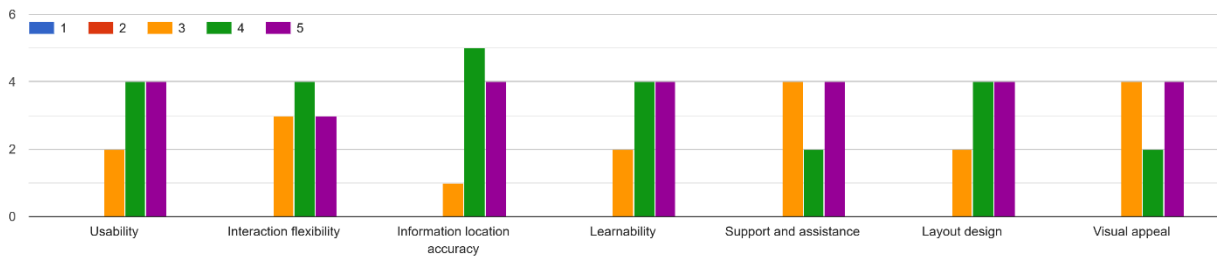
Upload Document - Specify properties - Choose printer:

Upload Document, Specify properties and Choose printer UI



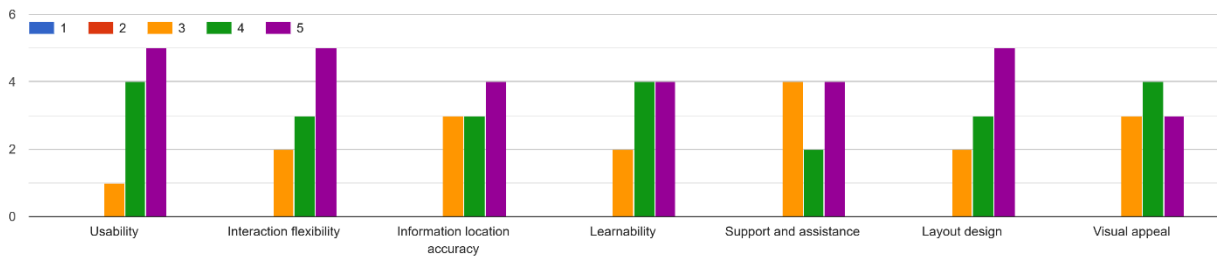
Student View log:

Student View Log UI



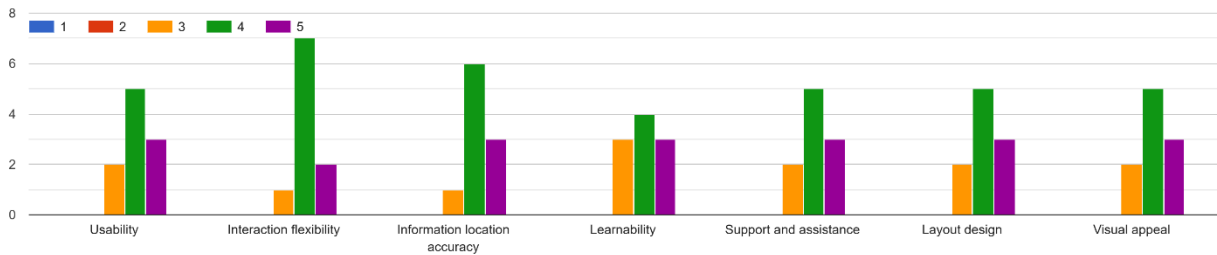
Student printing summary:

Student Summary UI



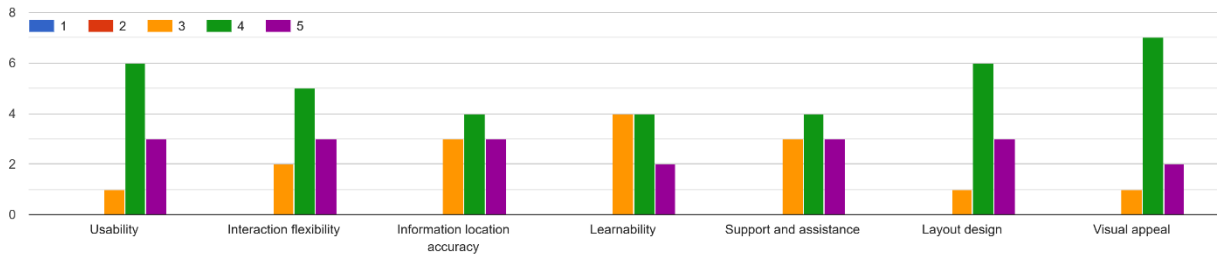
Manage printer:

SPSO Manage Printer UI



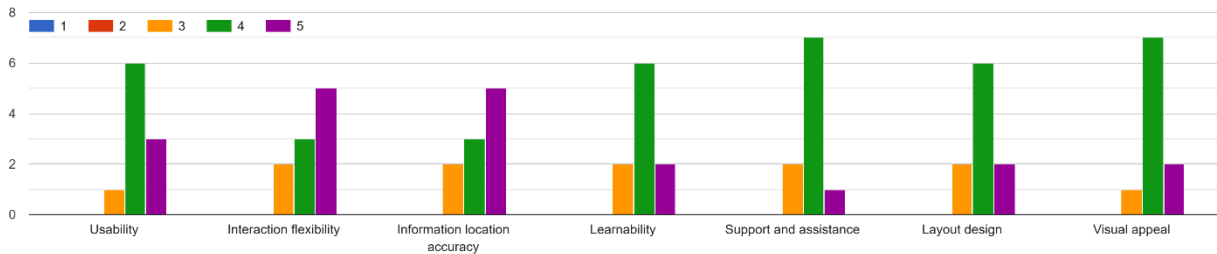
Manage configuration:

SPSO Manage configuration



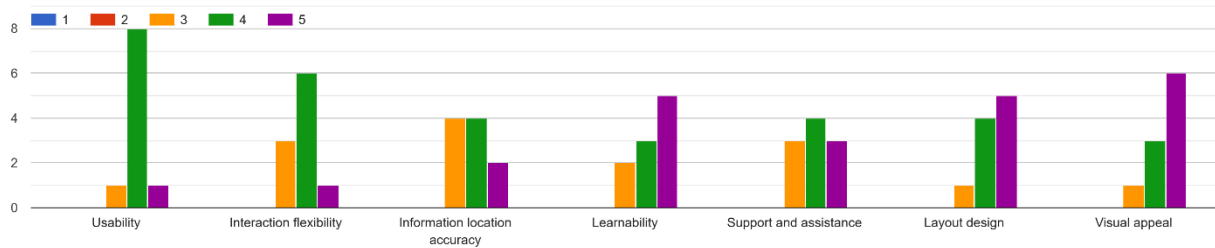
SPSO View student log:

SPSO view student log



SPSO View report:

SPSO view report



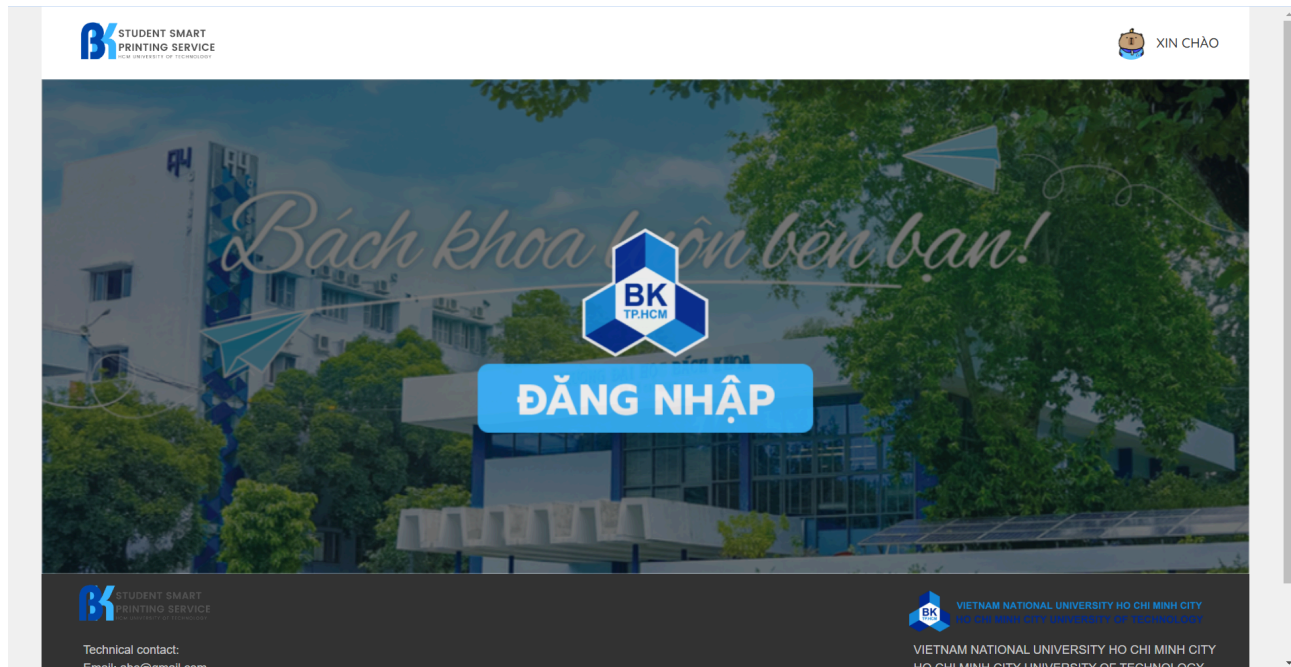
The feedback for testing our MVP1 ranges from 3 to 5 rate, which means from Neutral to Strongly agree. This indicates that users generally have a positive impression of our product.

b. Proposal for System Improvement

Changes	Explanation	Priority Level
Notify when successfully adding printer	Need a pop up to notify that the printer has successfully added to the database and refresh the list.	High
Close button for adding printer window	I want to return to the printer list from the adding-printer window.	Low
Showing the list	The printer list may be sorted and the active printers need to appear on the top.	Medium
Upload multiple file	Should be 1 file each time because it can affect the calculation of the pages, properties checking, ...	High
File Upload Feature Enhancement	Adding a preview section after file upload so users can confirm their uploaded content and avoid mistakes.	High
Printer printing log	There are should be a list of printing document for each printer	High

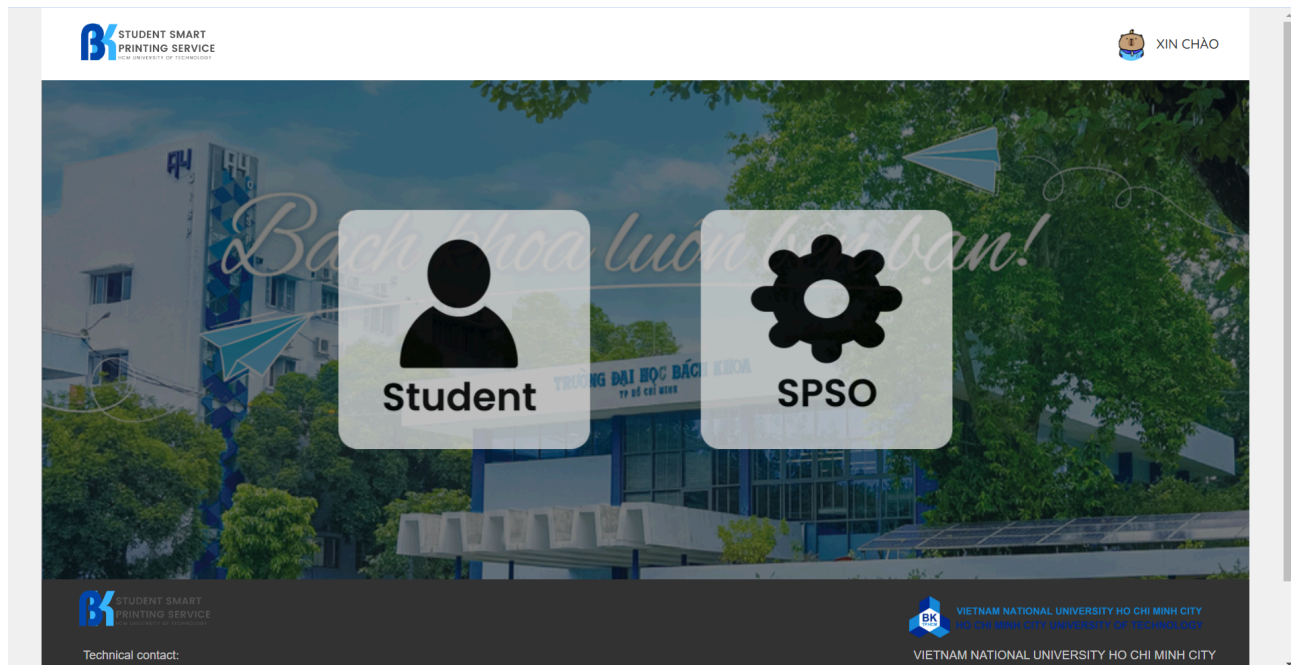
5. Task 5: Implementation – Sprint 2

5.1 Home page



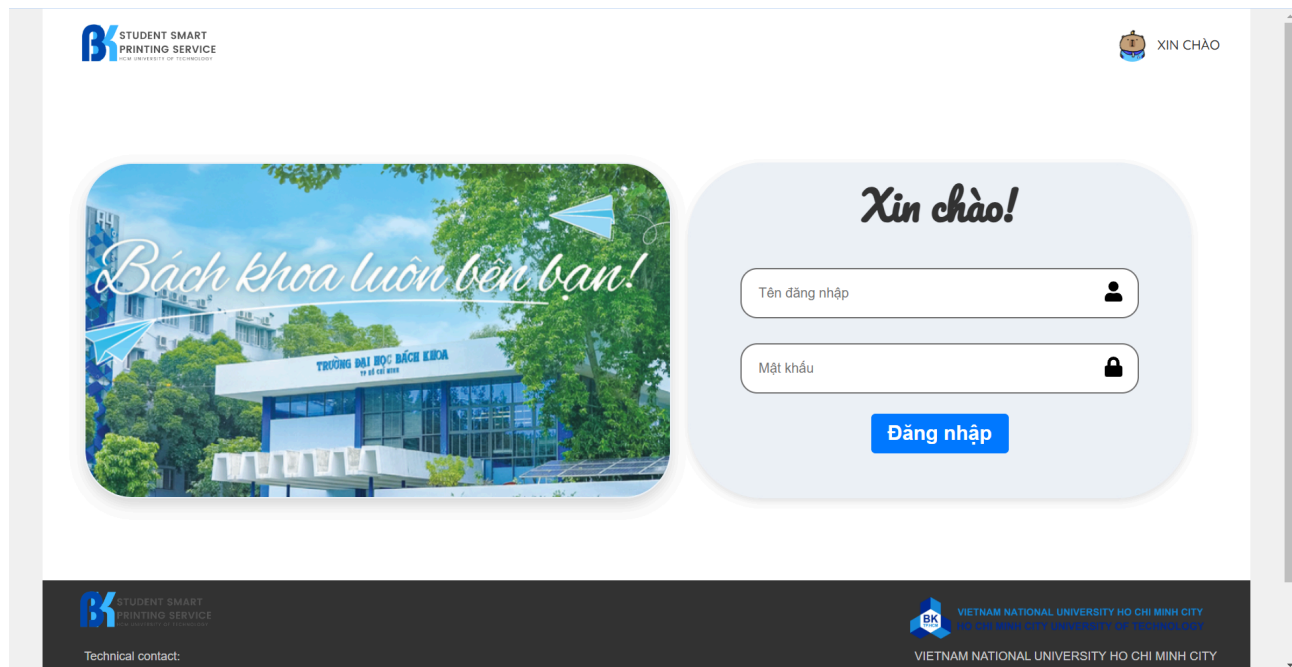
HCMUT_SPSS Home page

5.2 Login page



HCMUT_SPSS Choose user page

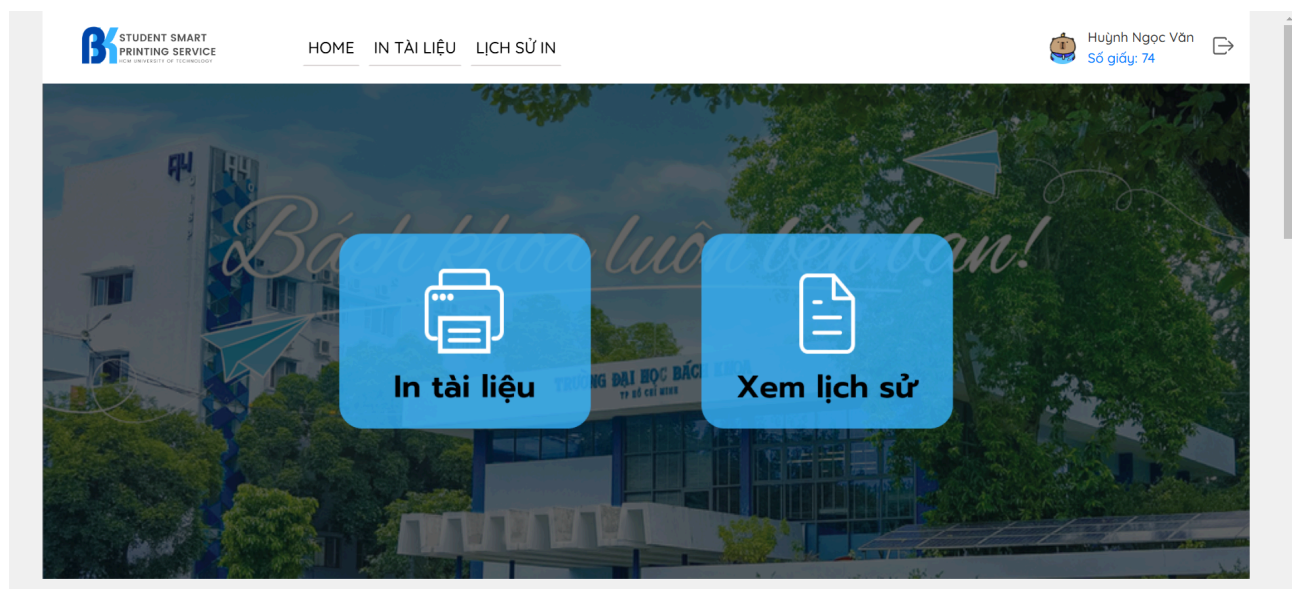
After selecting the login option on the home page, users must choose their role by selecting either 'Student' or 'SPSO' to proceed.



HCMUT_SPSS Login page

5.3 Student view

5.3.1 Student home page



HCMUT_SPSS Student home page

Upon selecting the “Student” role, the corresponding home page is displayed. The interface is divided into two main sections: the selection bar and the main content area, which offers two options — “*In tài liệu*” and “*Xem lịch sử*”.

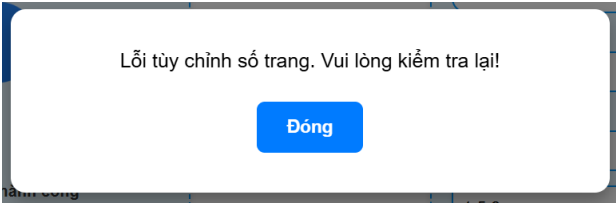
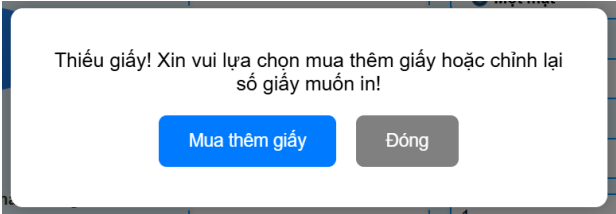
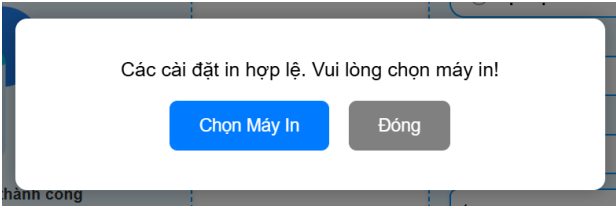
The left side of the selection bar features the user's avatar, username, and current page balance, providing a clear overview for users to easily monitor their balance.

5.3.2 Print document page

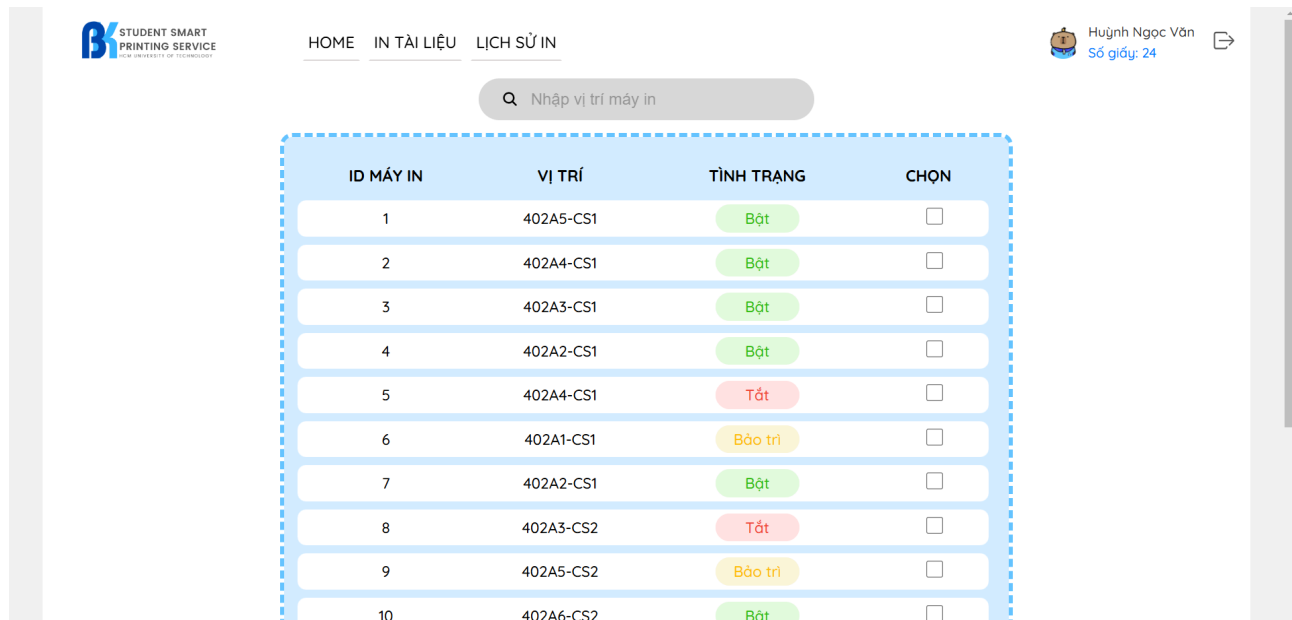
HCMUT_SPSS Student printing document page

Upon entering the printing section, the interface is divided into two parts: the right side is for uploading files, with specified permitted file types and size limits, while the left side allows users to select printing properties, including the side option, paper size, page range, and number of copies. Users must complete all fields to proceed to the next step. If any errors or invalid inputs occur, a corresponding popup will appear to notify the user.

Pop up message	Description
	pdf, docx, jpg, png Độ lớn tệp tin tối đa 50MB The file selected for upload must comply with the permitted file types and size limitations.

	Số trang Tùy chỉnh Nhập số trang. Ví dụ: 1-5, 8, 11-13 In the 'Số trang' input field, if the user chooses to custom input the pages for printing, the format must follow '1-5, 8, 11-13'.
	After entering all required fields, the system will calculate the paper needed to print the document. If the paper requirement exceeds the user's page balance, a popup will be displayed.
	“Buy more paper” feature
	If all the required inputs are valid, user choose “Chọn máy in” to move to the next step.

The “Choose Printer” interface displays a table of all printers in the system, as shown in the following figure.



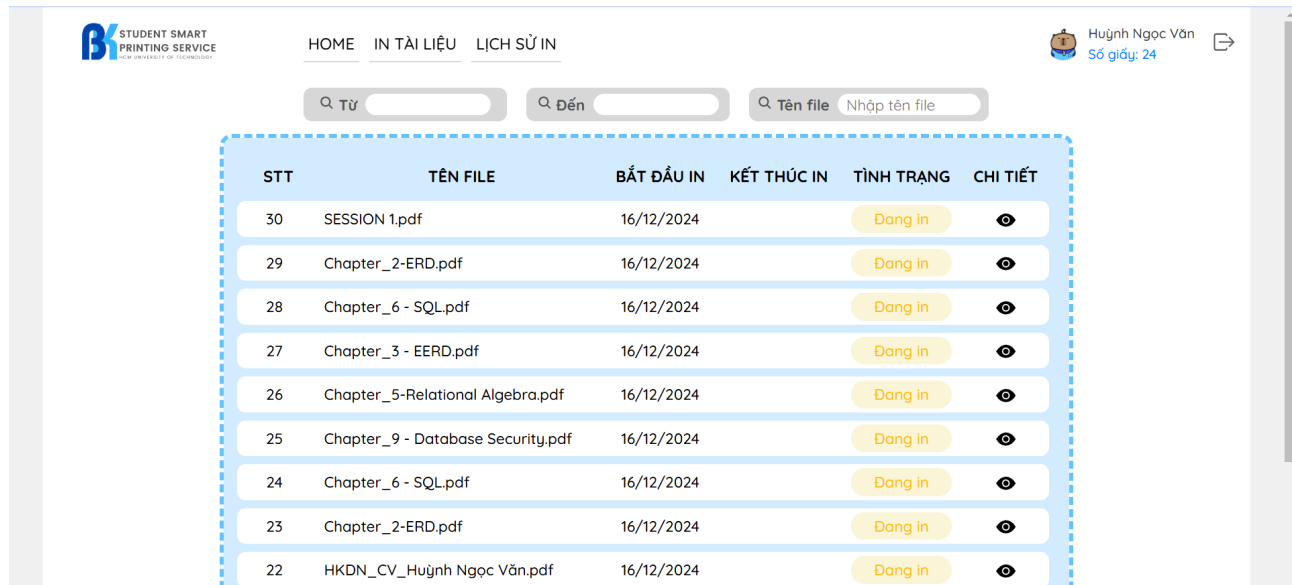
HCMUT_SPSS Choose printer page

Users can utilize the sort by column feature and the search bar to filter printers by location and then select a printer by marking the corresponding checkbox. Please note that users can only select printers with a “Bật” (“On”) status.

There is the list of popup message:

Pop up message	Description
	User do not choose a printer or the chosen printer is not in on status
	Printing request has been sent successfully

5.3.3 View log page



STT	TÊN FILE	BẮT ĐẦU IN	KẾT THÚC IN	TÌNH TRẠNG	CHI TIẾT
30	SESSION 1.pdf	16/12/2024		Đang in	
29	Chapter_2-ERD.pdf	16/12/2024		Đang in	
28	Chapter_6 - SQL.pdf	16/12/2024		Đang in	
27	Chapter_3 - EERD.pdf	16/12/2024		Đang in	
26	Chapter_5-Relational Algebra.pdf	16/12/2024		Đang in	
25	Chapter_9 - Database Security.pdf	16/12/2024		Đang in	
24	Chapter_6 - SQL.pdf	16/12/2024		Đang in	
23	Chapter_2-ERD.pdf	16/12/2024		Đang in	
22	HKDN_CV_Huỳnh Ngọc Văn.pdf	16/12/2024		Đang in	

HCMUT_SPSS Student view log page

In the “Xem lịch sử” section, users are presented with a table that displays essential information for all documents that have been sent for printing with the order of latest to oldest. This table provides a clear overview of each document, making it easy to track printing history.

Additionally, users can take advantage of sorting by column and the search bars to filter the records by specific time periods or file names, helping them find the relevant documents quickly.

For further details, users can simply click on the view icon on each row. This will open a popup that provides comprehensive information about the selected document.

Thông Tin Yêu Cầu In

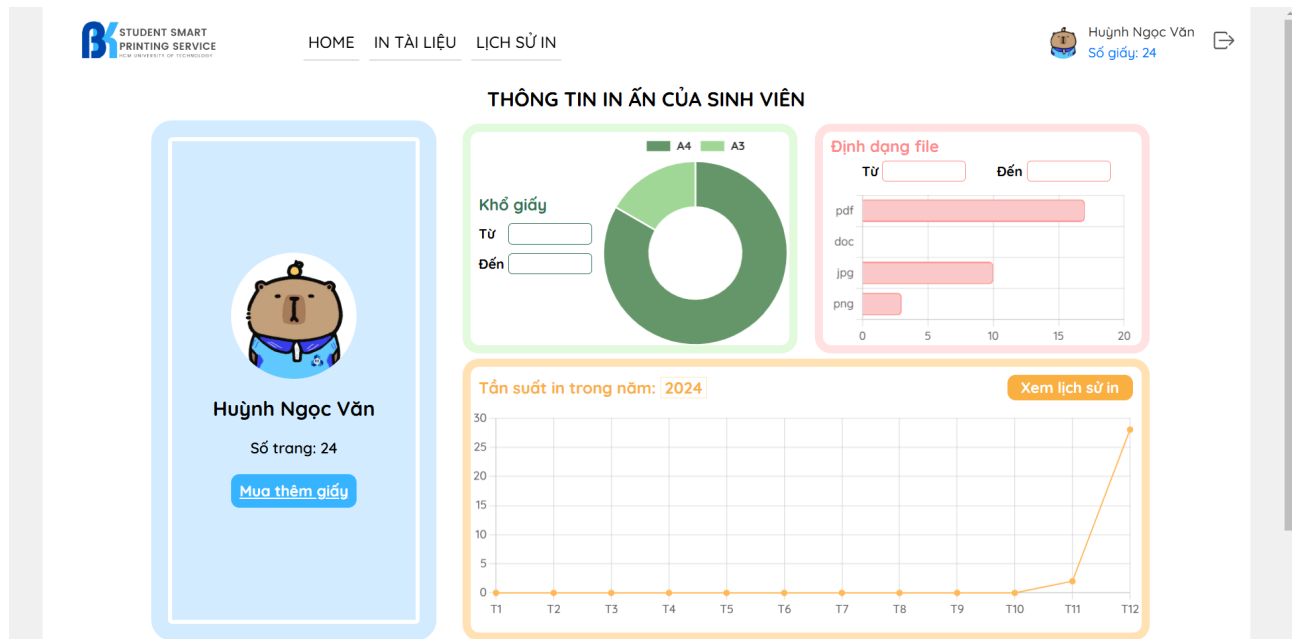
Tên file:	SESSION 1.pdf
ID máy in:	22
Ngày bắt đầu in:	16/12/2024
Ngày kết thúc in:	
Ngày nhận:	
Nơi nhận:	505B4-CS1
Trạng thái:	Đang in

Khổ giấy:	A4
Trang in:	1
In một/hai mặt:	Hai mặt
Số bản sao:	100

[ĐÓNG](#)

Detailed information of the printing request

5.3.4 View summary page

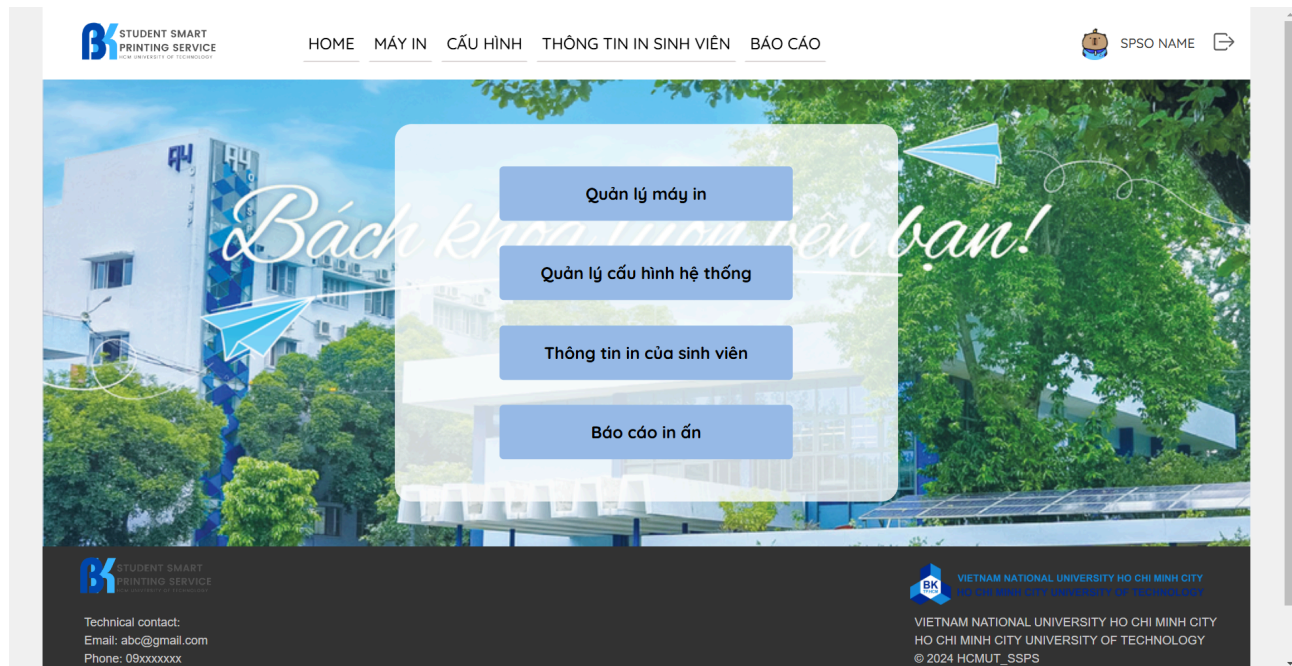


HCMUT_SPSS Student summary page

The website offers users a summary of their usage, including details on paper size, file type, and printing frequency throughout the year. Users can utilize the search bar to view statistics for a specific period, allowing them to easily analyze and track their usage over time.

5.4 SPSO view

5.4.1 SPSO home page



HCMUT_SPSS SPSO home page

Upon selecting the “SPSO” role, the corresponding home page is displayed. The interface is divided into two main sections: the selection bar and the main content area, which offers four options — “*Quản lý máy in*” (Manage printer), “*Quản lý cấu hình hệ thống*” (Manage configuration), “*Thông tin in của sinh viên*” (View student log) and “*Báo cáo in ấn*” (Report).

5.4.2 Mange printer page

a. Add printer

THÊM THÔNG TIN MÁY IN

Tên máy in:

Vị trí:

Địa chỉ IP:

Trạng thái: Bột

Số trang:

THÊM

HCMUT_SPSS Add printer page

In the “Add Printer” section, there is a form where users can enter the necessary information to add a printer to the system.

b. Modify printer's information

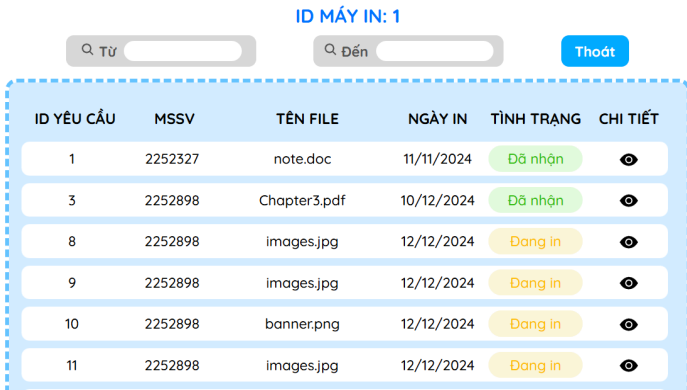
Q **THÊM MÁY IN**

ID MÁY IN	VỊ TRÍ	SỐ GIẤY	TÌNH TRẠNG	LỊCH SỬ IN	CÀI ĐẶT
1	402A5-CS1	500	Bột		
2	402A4-CS1	500	Bột		
3	402A3-CS1	110	Bột		
4	402A2-CS1	200	Bột		
5	402A4-CS1	105	Tắt		
6	402A1-CS1	300	Bảo trì		
7	402A2-CS1	200	Bột		
8	402A3-CS2	200	Tắt		
9	402A5-CS2	100	Bảo trì		
10	402A6-CS2	110	Bột		
11	402A7-CS1	102	Tắt		

Printer list of HCMUT_SPSS system

The interface displays a table of all printers in the system, users can utilize the sort by column feature and the search bar to filter printers.

There are some key features that the interface provide:

User interface	Description
	Modify printer status: Users can click on the “setting” icon in a row to access the modification interface for the corresponding printer.
	Printer log: Users can click the “view” icon to access the printing history of a specific printer. This interface includes a feature that allows users to filter the history by date, making it easier to review specific time periods.

Users can also refill the printer’s paper by clicking on the icon next to the paper count in the “Số giấy” column. In this case, the system will automatically update the paper count to 500.

5.4.3 Manage configuration page

HCMUT_SPSS SPSO manage configuration page

For the “Manage configuration” option, there is a form where users can enter the necessary information to modify the configuration of the system.

5.4.4 View student log page

MSSV	ID MÁY IN	TÊN FILE	NGÀY IN	TÌNH TRẠNG	CHI TIẾT
2252898	19	Giấy kê ô A4.pdf	09/11/2024	Đã nhận	👁
2252898	7	allexam.pdf	11/11/2024	Đã nhận	👁
2252898	1	Chapter3.pdf	10/12/2024	Đã nhận	👁
2252898	2	Quiz#01. Intro.pdf	10/12/2024	Chưa nhận	👁
2252898	2	1.1_1.2.pdf	11/12/2024	Đang in	👁
2252898	3	Feature Splatting.pdf	11/12/2024	Đang in	👁
2252898	19	Software-Engineering.pdf	12/12/2024	Đang in	👁
2252898	1	images.jpg	12/12/2024	Đang in	👁
2252898	1	images.jpg	12/12/2024	Đang in	👁
2252898	1	banner.png	12/12/2024	Đang in	👁
2252898	1	images.jpg	12/12/2024	Đang in	👁

HCMUT_SPSS View student log page

In the “View student log” section, users are presented with a table that displays essential information for all documents that have been sent for printing. This table provides a clear overview of each document, making it easy to track printing history.

Additionally, users can take advantage of sorting by column and the search bars to filter the records by specific time periods, student id or printer id, helping them find the relevant documents quickly.

For further details, users can simply click on the view icon on each row. This will open a popup that provides comprehensive information about the selected document.

Thông Tin Yêu Cầu In	
MSSV:	2252898
Tên sinh viên:	Huỳnh Ngọc Văn
Tên file:	Giấy kê ô A4.pdf
ID máy in:	19
Ngày bắt đầu in:	09/11/2024
Ngày kết thúc in:	09/11/2024
Ngày nhận:	09/11/2024
Nơi nhận:	402A2-CS1
Trạng thái:	Đã nhận

Khổ giấy:	A4
Trang in:	Tất cả trang
In một/hai mặt:	Hai mặt
Số bản sao:	1

ĐÓNG

Detailed information of the printing request

5.4.5 Report page

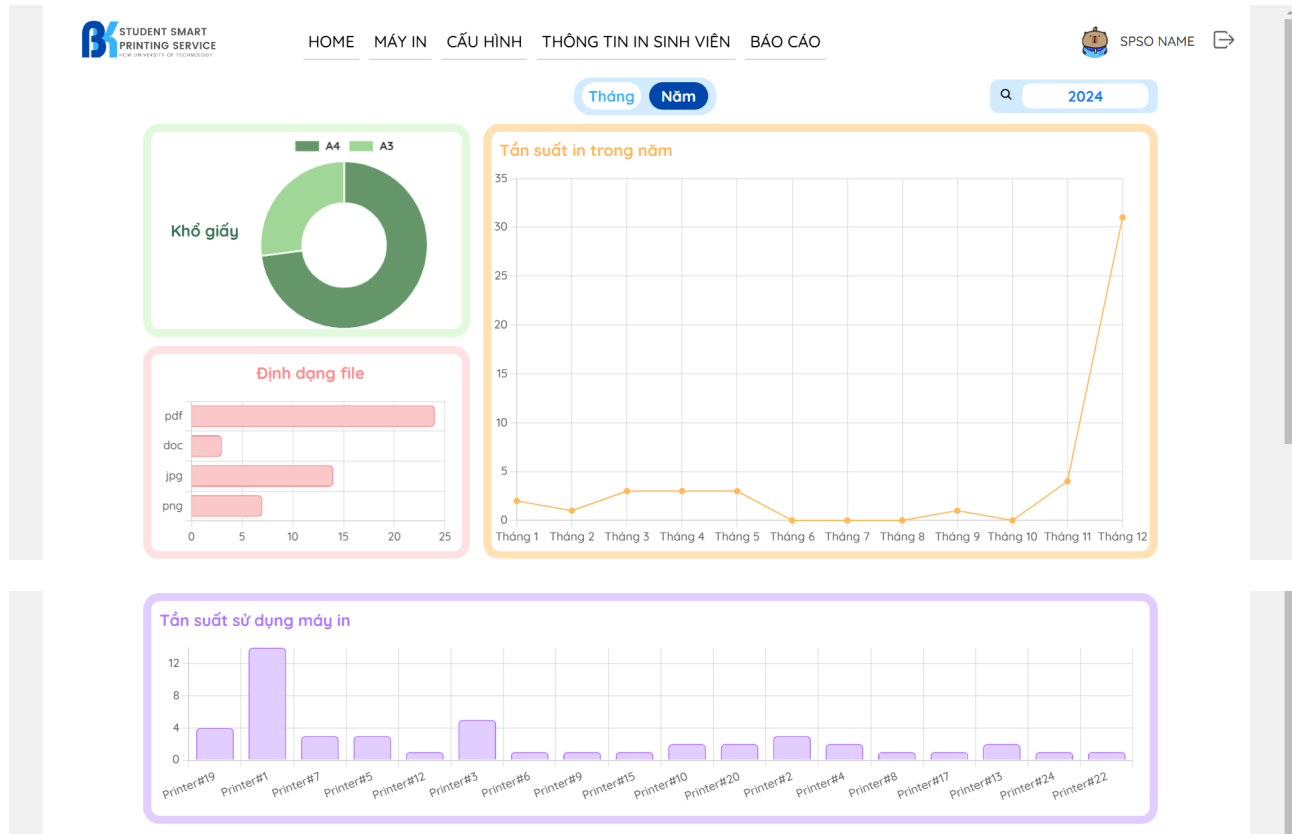
a. Monthly report



HCMUT_SPSS SPSO monthly report

The website offers users a summary of usage of the system in a particular month of a year, including details on paper size, file type, printer usage frequency and printing frequency.

b. Annual report



HCMUT_SPSS SPSO annual report

The website offers users a summary of usage of the system in a particular year, including details on paper size, file type, printer usage frequency and printing frequency.